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Integrated scientometric mapping and chemometric profiling of *Siparuna guianensis* essential oil: chemical variability, antioxidant and biological activity

Danylo Duran Silva Santos · Lucas Villar Pedrosa Silva Pantoja · Joana Julia dos Santos Moraes · Eliza de Jesus Barros dos Santos · Ravendra Kumar · Mateus Machado Lima · Karyme do Socorro de Souza Vilhena · Antônio Maia de Jesus Chaves Neto · Brenda Costa da Conceição · Marta Eduarda Oliveira Barbosa · Eloise Karoline Serrão Cardoso · Cristiane Socorro Ferraz Maia · Enéas Andrade Fontes-Junior · Mozaniel Santana de Oliveira

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Abstract *Siparuna guianensis* is a neotropical aromatic species widely used in traditional medicine whose essential oil exhibits diverse biological activities. However, current evidence remains fragmented and strongly influenced by chemical variability across populations and extraction conditions. This study integrates scientometric mapping, multivariate chemometric analysis, and critical pharmacological review to clarify the relationship between chemical composition and biological activity of *S. guianensis* essential oil. Publications indexed in Web of Science were analyzed to identify research trends, collaboration networks, and thematic focus. Reported chemical composition data were subjected to principal

D. D. S. Santos · L. V. P. S. Pantoja · J. J. dos Santos Moraes · M. E. O. Barbosa · E. K. S. Cardoso · C. S. F. Maia · E. A. Fontes-Junior · M. S. de Oliveira (✉)
Programa de Pós-Graduação Em Ciências Farmacêuticas, Universidade Federal Do Pará, Belém, PA 66075-900, Brazil
e-mail: mozaniel.oliveira@yahoo.com.br

D. D. S. Santos · J. J. dos Santos Moraes · B. C. da Conceição · M. E. O. Barbosa · E. K. S. Cardoso · C. S. F. Maia · E. A. Fontes-Junior · M. S. de Oliveira
Laboratório de Farmacologia da Inflamação E Do Comportamento, Instituto de Ciências da Saúde, Universidade Federal Do Pará, Belém, PA 66075-900, Brazil

L. V. P. S. Pantoja · B. C. da Conceição · C. S. F. Maia · E. A. Fontes-Junior
Programa de Pós-Graduação Em Farmacologia E Bioquímica, Universidade Federal Do Pará, Belém, PA 66075-900, Brazil

E. de Jesus Barros dos Santos
Postgraduate Program in Biological Sciences - Tropical Botany, Universidade Federal Rural da Amazônia, and Museu Paraense Emílio Goeldi, Av. Terra Firme, Belém, Pará 66075- 900, Brazil

R. Kumar
Department of Chemistry, College of Basic Sciences and Humanities, G.B. Pant, University of Agriculture and Technology, Pantnagar, U.S. Nagar 263145, Uttarakhand, India

M. M. Lima · K. do Socorro de Souza Vilhena
Department of Chemistry, Institute of Exact and Natural Sciences, Universidade Federal Do Pará, Augusto Corrêa Rua, S/N, Guamá, Belém, Pará 66075-900, Brazil

A. M. de Jesus Chaves Neto
Laboratory of Preparation and Computation of Nanomaterials, Faculty of Physics, Federal University of Pará, Rua Augusto Correia, 01, Belém, Pará, Brazil

component analysis and hierarchical clustering to define chemotype patterns and associate them with biological outcomes. The literature reveals a highly centralized and chemistry-driven research field dominated by vector-control and pest-management studies, with limited mechanistic pharmacology in mammalian systems. Chemometric results demonstrate pronounced chemical polymorphism, separating monoterpene-rich and sesquiterpene-rich chemotypes. Monoterpene profiles, particularly those dominated by β-myrcene and aliphatic ketones, are mainly linked to insecticidal and repellent effects, whereas sesquiterpene-rich oils containing germacrene D, curzerenone, spathulenol, and α-bisabolol are more frequently associated with antimicrobial, antiparasitic, enzyme-inhibitory, and antitumoral activities. Despite promising results, most studies remain preliminary. Pharmacological translation will require chemotype standardization, mechanistic validation, safety profiling, and advanced formulation strategies to enable reproducible and clinically relevant applications.

Keywords Chemotype variability · Multivariate statistics · Principal component analysis · Pharmacological screening · Natural product bioactivity

Introduction

Medicinal and aromatic plants (MAPs), traditionally used for therapeutic purposes, are currently important raw materials for the production of natural compounds with a wide range of applications, including cosmetic, food, biomedical, and industrial products (Oliveira 2022). Among their derivatives, essential oils (EOs) stand out as complex mixtures of bioactive secondary metabolites, which have attracted considerable interest due to their effective therapeutic properties, supported by ethnopharmacological knowledge and traditional uses, as well as by their favorable chemical compositions(Fierascu et al. 2021; Pires et al. 2021; Dias et al. 2021; Ferreira et al. 2024).

Numerous health-beneficial biological properties associated with EOs have been reported, including their use in the prevention and management in the oxidative stress, as well as cardiovascular, infectious, respiratory, neoplastic, and inflammatory diseases (Fierascu et al. 2021; Pinto et al. 2021). However, these biological activities vary according to their specific chemical compositions, which may be influenced by environmental, genetic, and methodological factors. This variability highlights the pronounced chemical diversity of EOs and underscores the importance of characterizing their different chemotypes (de Souza et al. 2018; Burneo et al. 2021; Botelho et al. 2025).

Currently, several plant groups have been investigated due to their chemical diversity, including species of the genus *Siparuna*. Belonging to the family Siparunaceae, this neotropical genus is distributed from Mexico to Paraguay and comprises aromatic trees or shrubs characterized by simple, opposite leaves, inconspicuous flowers, and multiple small globose fruits with a distinctive pungent odor (Hall and Meirelles 2017). The genus includes approximately 60 species, of which about 20 occur in Brazil, except in the states of Santa Catarina and Rio Grande do Sul, with the Amazon region harboring the highest diversity, accounting for 19 species (Brunassi et al. 2022; Peixoto 2026).

Although they are neither ornamental nor fruit-bearing plants, some *Siparuna* species exhibit significant economic, therapeutic, and phytochemical potential due to their ability to produce EOs and their extensive use in traditional medicine by different Indigenous and riverside communities across Central and South America (Brunassi and Lírio 2025). Leaves, fruits, and bark are commonly used by traditional healers in spiritual and religious rituals, as well as to treat conditions ranging from snake bites, wounds, fevers, and colds to gastrointestinal, infectious, respiratory, and rheumatic diseases(Burneo et al. 2021; Morocho et al. 2023; Brunassi and Lírio 2025).

More recent studies have described the main chemical groups most frequently reported in EOs of the genus, namely monoterpenes and sesquiterpenes, present in both hydrocarbon and oxygenated forms. In the EO of *Siparuna cymosa*, the presence of α-bisabolol, *p*-cymen-9-ol, and spathulenol was reported as major constituents, exhibiting moderate cytotoxic activity against leukemic tumor cells in vitro (Oliveira et al. 2020). Spathulenol was also identified as the major compound in *S. thecaphora*, followed by 2-tridecanone, α-copaene, α-cadinol, and δ-cadinene; in this species, the EO showed strong bacteriostatic activity (Vila et al. 2002).

Studies on *S. schimpffi* and *S. aspera* identified germacrene D, bicyclogermacrene, and α-pinene as the main constituents, whereas *S. macrotepala* presented δ-cadinene instead of α-pinene; the EOs of the latter two species exhibited relevant antioxidant and antimicrobial activities (Paco Fernando Noriega Rivera, Alexandra Guerrini and Tsamaraint 2014; Noriega et al. 2019). In the EO obtained from the leaves and fruits of *S. muricata*, high concentrations of α-pinene, limonene, β-acorenol, β-pinene, and camphene were reported, demonstrating moderate antioxidant activity and strong antimicrobial and anticholinesterase effects. Comparatively, *S. guianensis* also exhibited anticholinesterase activity, with an EO rich in shyobunone and its derivatives (45.5% of the total EO), reinforcing the neuroprotective potential of the genus (Martins et al. 2021; Brunassi and Lírio 2025).

Among the Siparunaceae, the species *Siparuna guianensis*, popularly known as "negramina" and "capitiú," stands out for being frequently used as a model in phytochemical and pharmacological studies, mainly due to its chemical richness, reported biological properties, and wide geographic distribution, occurring across all Brazilian biomes (Martins et al. 2021; Brunassi and Lírio 2025). Several biological activities have already been investigated for this species, including antimicrobial (de Jesus et al. 2020), antiparasitic (Andrade et al. 2016), and anthelmintic activities (Carvalho et al. 2020), among others; however, the most frequently reported properties of its EO are its insecticidal and repellent potentials (Andrade et al. 2015; Ferreira et al. 2020; Montaño-Campaz et al. 2025).

Despite its extensive traditional use and growing scientific interest, studies involving the essential oil of *S. guianensis* remain limited, scattered, and poorly integrated, particularly with respect to linking phytochemical variability to biological outcomes beyond in vitro models. In this context, the aim of this study was to integrate scientometric analysis, multivariate chemometric profiling, and critical literature review to systematically organize the existing knowledge on the chemical composition, antioxidant capacity, and biological activities of *S. guianensis* essential oil. By combining these complementary approaches, this study provides a structured framework to identify research trends, define chemotype patterns, and establish associations between chemical variability and biological activity. This integrative strategy is expected to support the rational interpretation of fragmented data, guide future experimental design, and facilitate the development of more reproducible and translational studies, thereby contributing to advancing the scientific understanding and pharmacological potential of this species..

Methodology

Bibliometric procedures

A bibliometric analysis was performed following procedures previously established by our research group (Maia et al. 2023; Conceição et al. 2023; Silva et al. 2025). The Web of Science Core Collection (WoS-CC) database was used as the primary data source to retrieve publications related to *Siparuna guianensis* essential oil and its biological and antioxidant properties. The literature search was performed on 06 October 2026, using a predefined search strategy ([*Siparuna OR "Siparuna guianensis" OR "S. guianensis" OR Negramina] AND ["Essential oil" OR "Volatile oil"]*). Eligible documents included only original research articles addressing the chemical composition and biological activities of *S. guianensis* essential oil, with no language restrictions applied. Review articles, conference papers, editorials, letters, proceedings, ethnobotanical studies, articles not freely available, and publications lacking biological evaluation were excluded. After establishing the inclusion criteria, two independent researchers screened titles and abstracts. Disagreements regarding article eligibility were resolved by consultation with a third senior researcher.

Bibliometric indicators were extracted from WoS-CC, including article title, publication year, authors, number of citations, journals, countries, institutions, and author keywords. Network visualization and quantitative mapping were performed using VOSviewer software (version 1.6.16). In the generated networks, nodes represent authors, institutions, or keywords, while links indicate co-authorship, inter-institutional collaboration, or keyword co-occurrence; node size is proportional to publication or citation frequency, and link strength reflects the intensity of collaboration or thematic association. Institutional productivity and collaboration patterns were also analyzed to identify core research hubs. Total link

strength values were used to quantify collaborative relationships among authors and institutions. The geographical distribution of publications was illustrated using MapChart (https://mapchart.net/) (Fig. 1).

Multivariate analysis methodology (PCA and HCA)

The chemical composition data of *Siparuna guianensis* (Aubl.), compiled from different studies reported in the literature, were organized into a data matrix in which rows represented the samples and columns corresponded to the identified chemical constituents, expressed as relative percentage (%). Only compounds reported in the selected studies were included in the multivariate analysis. Prior to the statistical analyses, the data were subjected to autoscaling standardization (mean-centered and variance-scaled to unit variance) in order to eliminate the influence of differences in variable magnitude and to ensure an equivalent contribution of all chemical constituents.

Principal Component Analysis (PCA) was applied as an exploratory method to reduce data dimensionality and to identify patterns of similarity and variability among samples, considering variations associated with different years of collection and reported chemical profiles. The principal components were selected based on eigenvalues greater than 1 (Kaiser criterion) and cumulative explained variance, with emphasis on those components accounting for the largest proportion of the total variance. The PCA results were interpreted using score plots, which allowed visualization of sample distribution and grouping, and loading plots, which were used to identify the chemical compounds most responsible for sample discrimination. Hierarchical Cluster Analysis (HCA) was performed as a complementary technique to PCA in order to assess the similarity among samples and to confirm the clustering patterns observed. HCA was conducted using

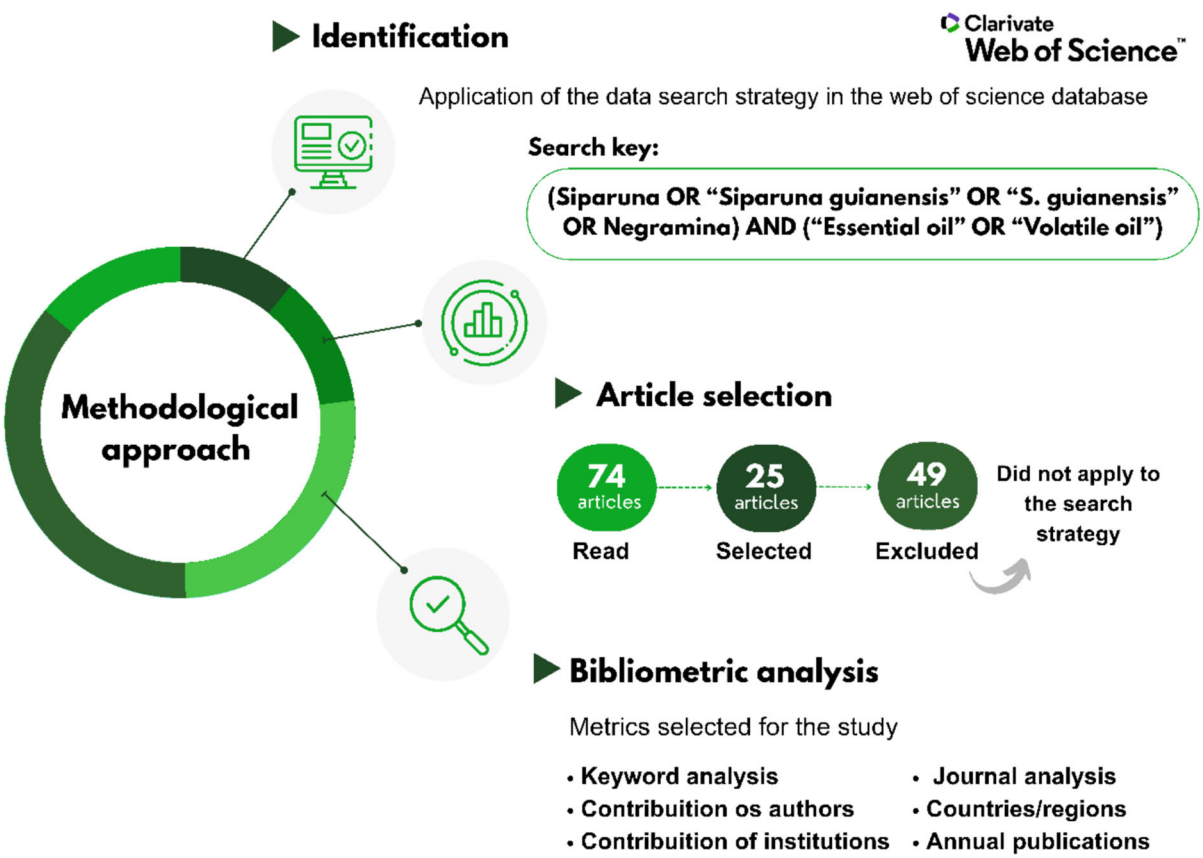


Fig. 1 Methodological strategy applied to perform bibliometric analysis

Euclidean distance as the dissimilarity measure and the complete linkage method as the clustering criterion. The results were expressed as dendrograms, allowing visualization of similarity relationships among samples and identification of groups with similar chemical profiles. All statistical analyses were carried out using Minitab® version 22 (free version), adopting an exploratory approach suitable for review studies based on secondary data obtained from different bibliographic sources.

Literature review

After bibliometric filtering and article selection, additionally, the synonyms of the species' scientific name were verified using informations from the following online databases: (Plants of the World Online) "The Plant List", "World Flora Online" and "Flora do Brasil" (Peixoto 2026), which identified the following synonyms for the species: *"Siparuna discolor"; "Siparuna panamensis"; "Angelina divergentifolia"; "Citrosma camporum"; "Citrosma discolor"; "Citrosma guianensis"; "Citrosma guianensis f. divergentifolia"; "Citrosma guianensis f. nuda"; "Citrosma guianensis f. nuda"; "Citrosma oligocarpa"; "Siparuna foetida"; "Siparuna guianensis var. divergentifolia"; "Siparuna guianensis var. glabrescens"; "Siparuna guianensis var. longifólia"; "Siparuna guianensis var. nitens"; "Siparuna savanicola"; "Siparuna archeri"; "Siparuna arianeae"; "Siparuna cavalcantei"; "Siparuna duckeana"; "Siparuna itacaiunensis"; "Citriosma guianensis"; "Citrosma guianensis var. nuda";"Siparuna camporum"*. A narrative literature review was then conducted to integrate and contextualize the experimental evidence regarding the essential oil of *S. guianensis*, with primary emphasis on reported biological activities. The full texts of all included articles were carefully examined, focusing on chemical composition, major constituents, and biological activity.

Results and discussion

Bibliometric results

The search strategy applied in WoS-CC retrieved 74 documents related to *Siparuna guianensis*. After applying the predefined inclusion criteria, 25 articles were selected for detailed qualitative and quantitative evaluation. This marked reduction indicates that, although *S. guianensis* has been extensively investigated from botanical and phytochemical perspectives, only approximately 34% of the available literature establishes a direct association between volatile composition and biological function. This finding reveals a fragmented research landscape characterized by abundant chemical profiling but comparatively limited bioprospecting of biological activities.

The earliest and most cited study was published in 2013 and investigated the chemical composition and antioxidant properties of *S. guianensis* essential oil; however, no antioxidant activity was detected (Andrade et al. 2013). In contrast, the most recent publication (2025) employed an integrated approach combining phytochemical characterization, in silico molecular docking, and in vitro bioassays to demonstrate larvicidal activity, highlighting acetylcholinesterase inhibition as a potential mechanism of action (Montaño-Campaz et al. 2025) (Table 1). The analysis of the selected articles indicates that research on *S. guianensis* essential oil is predominantly centered on insecticidal and vector-control applications, particularly against *Aedes aegypti* and agricultural pests, which also account for the most highly cited publications. In contrast, studies approaching antimicrobial, antioxidant, antiparasitic, or antiviral activities generally exhibit lower citation impact and often report only moderate efficacy which highlights the peripheral role of *S. guianensis* in these research domains.

Phytochemical data reveal pronounced intraspecific variability, with β-myrcene, α-bisabolol, spathulenol, curzerenone, germacrene D, γ-elemene, and shyobunone derivatives emerging as recurrent major constituents (a comprehensive phytochemical characterization is addressed in Sect. "Chemical compositions Siparuna guianensis"). Biological activity appears to be strongly chemotype-dependent, as evidenced by studies demonstrating superior acaricidal and anthelmintic effects in α-bisabolol-rich oils and consistent insecticidal performance in β-myrcene, dominant profiles. More recent investigations increasingly integrate molecular docking, enzyme inhibition assays, and membrane permeability studies, providing more sophisticated biological assays. In addition to this, a growing number of publications incorporate formulation strategies such as nanoemulsions and

Table 1 Selected articles on *Siparuna guianensis* essential oil and its biological properties

Authors/ Year	Article Title	DOI	Number of citations (WoS)	Brief description
Montaño-Campaz et al. (2025)	Siparuna gesnerioides and *Siparuna guianensis* Essential Oils in Aedes aegypti Control: Phytoanalysis, Molecular Insights for Larvicidal Activity and Selectivity to Non-Target Organisms	https://doi.org/10.3390/plants14091322	1	This study employed an in silico-guided approach combining chemical profiling, molecular docking, and biological assays. The essential oil was characterized by γ-elemene and germacrene D as major constituents. Docking analyses predicted strong binding affinity of both compounds to Aedes aegypti acetylcholinesterase. These findings were supported by larvicidal assays, which demonstrated pronounced toxicity against *A. aegypti* larvae. These results suggest acetylcholinesterase inhibition as a potential mechanism underlying the observed larvicidal activity, likely leading to enhanced cholinergic tone
Pino-Benitez et al. (2024)	Repellent and insecticidal activity of four essential oils from plants recolleted in Choco-Colombia against *Tribolium castaneum*	https://doi.org/10.37360/blacpma.24.23.4.38	1	This study addressed the impact of agricultural pests on food production by evaluating the insecticidal potential of *Siparuna guianensis* essential oil against *Tribolium castaneum* larvae. The essential oil exhibited significant larvicidal activity, demonstrating both protective and lethal effects, and highlighting its potential as a natural alternative for pest management in stored-product systems
Santos et al. (2024)	Intraspecific Chemical Variability and Antioxidant Capacity of *Siparuna guianensis* Aubl. Essential Oil from Brazil	https://doi.org/10.3390/horticulturae10070690	1	This study investigated the intraspecific chemical variability of *Siparuna guianensis* essential oils obtained from different plant samples. Despite marked variation in overall composition, spathulenol was consistently detected across all oils at relatively high levels, suggesting its potential as a chemical marker. However, no significant antioxidant activity was observed
Barbosa et al. (2023)	*Siparuna guianensis* Essential Oil Antitumoral Activity on Ehrlich Model and Its Effect on Oxidative Stress	https://doi.org/10.1002/cbdv.202301120	6	This study reported, for the first time, the antitumor potential of *S. guianensis* essential oil. Chemical characterization identified curzerenone, drimenol, and spathulenol as major constituents. The essential oil significantly reduced tumor cell viability, supporting its cytotoxic activity. Beyond this, in vivo experiments demonstrated antioxidant effects, as evidenced by reduced hepatic lipid peroxidation and increased antioxidant markers, indicating systemic protective activity
Santos et al. (2023)	Chemical Composition and antioxidant activity of Essential Oil from *Schinus terebinthifolius* and *Siparuna guianensis* Leaves	https://doi.org/10.21577/1984-6835.20220119	1	In this study, the authors characterized the phytochemical profile of *Siparuna guianensis* essential oil obtained from plants collected in southern Tocantins. β-Myrcene was identified as the major constituent, and the oil exhibited antioxidant activity
Costa et al. (2022)	Molecular Docking in silico Analysis of Brazilian Essential Oils Against Host Targets and SARS-CoV-2 Proteins	https://doi.org/10.21577/0103-5053.20220043	4	In this computationally guided study, the authors first characterized the phytochemical profile of *Siparuna guianensis* essential oil and identified shyobunone derivatives as the predominant terpenoids. Subsequently, molecular docking analyses indicated that most phytocompounds exhibited high binding affinity toward SARS-CoV viral proteins, suggesting a potential antiviral activity

Table 1 continued

Authors/ Year	Article Title	DOI	Number of citations (WoS)	Brief description
Diniz et al. (2022)	Chemical composition of essential oils of different *Siparuna guianensis* chemotypes and their acaricidal activity against Rhipicephalus microplus (Acari: Ixodidae): influence of α-bisabolol	https://doi.org/10.1080/01647954.2021.2009910	6	The authors evaluated the chemical composition of essential oils from different *Siparuna guianensis* chemotypes and assessed their acaricidal activity against Rhipicephalus microplus. They identified α-bisabolol as the major compound in two samples and germacrene D in another. Chemotypes rich in α-bisabolol exhibited the highest larvicidal and adulticidal activities, indicating that chemical variability directly influences acaricidal efficacy
Souza et al. (2021)	*Siparuna guianensis* Controls Toxoplasma gondii Infection In Vitro	https://doi.org/10.1007/s43450-021-00194-x	1	The authors evaluated *Siparuna guianensis* essential oil and its major compound, α-bisabolol, against Toxoplasma gondii. Cytotoxicity assays in fibroblast cells showed that the essential oil was considerably toxic at 500 µg/mL, whereas α-bisabolol at the same concentration did not significantly affect cell viability, highlighting the complexity and potential toxicity of the oil mixture. Regarding antiprotozoal activity, α-bisabolol exhibited superior efficacy compared to the whole oil, suggesting its potential as a promising candidate for further studies on T. gondii infection
Martins et al. (2021)	Impact on cholinesterase-inhibition and in silico investigations of sesquiterpenoids from Amazonian *Siparuna guianensis* Aubl	https://doi.org/10.1016/j.saa.2021.119511	10	The authors analyzed the essential oil of *Siparuna guianensis* and identified sesquiterpenoids, particularly shyobunone and its derivatives, as the major constituents. The biological activity evaluated was in vitro anticholinesterase activity, with the essential oil exhibiting significant inhibitory effects. To elucidate the mechanism of action, the authors employed spectrophotometry, fluorescence quenching, 1H NMR titration, and molecular docking analyses, demonstrating that the major compounds interact with key binding sites of acetylcholinesterase, supporting enzyme inhibition as the underlying mechanism
Moura et al. (2021)	Cassava starch-based essential oil microparticles preparations: Functionalities in mosquito control and selectivity against non-target organisms	https://doi.org/10.1016/j.indcrop.2021.113289	15	In this study, the authors applied a pharmaceutical technology approach using cassava starch-based microencapsulation to improve the delivery of *Siparuna guianensis* essential oil and evaluate its larvicidal activity against Aedes aegypti. Phytochemical analysis identified the monoterpene β-myrcene as the major constituent. The microencapsulated essential oil effectively induced larval mortality, while toxicity assessment in Danio rerio revealed no significant adverse effects, supporting both efficacy and preliminary safety of this delivery system

Table 1 continued

Authors/ Year	Article Title	DOI	Number of citations (WoS)	Brief description
Santana de Oliveira et al. (2020)	Chemical Composition, Antimicrobial Properties of *Siparuna guianensis* Essential Oil and a Molecular Docking and Dynamics Molecular Study of its Major Chemical Constituent	https://doi.org/10.3390/molecules25173852	51	In this study, the authors evaluated the antimicrobial activity of *Siparuna guianensis* essential oil against *Streptococcus mutans*, *Enterococcus faecalis*, *Escherichia coli*, and *Candida albicans*. The oil inhibited all tested microorganisms, showing particularly strong effects against *S. mutans* and *C. albicans*. Phytochemical analysis identified curzerene, γ-elemene, germacrene D, trans-β-elemenone, and atractylone as the main constituents. Molecular docking further suggested atractylone (18%), the major compound, as a key contributor to activity, demonstrating stable and energetically favorable interactions with catalytic sites of target proteins
de Jesus et al. (2020)	Antimicrobial Potential of Essential Oils from Cerrado Plants against Multidrug-Resistant Foodborne Microorganisms	https://doi.org/10.3390/molecules25143296	17	In this study, the authors screened essential oils from several Cerrado plant species, including *Siparuna guianensis*, for antibacterial activity against foodborne pathogens. *S. guianensis* oil exhibited only moderate inhibitory effects compared with the other evaluated oils and, therefore, was not selected for subsequent phytochemical characterization
Ferreira et al. (2020)	Nano-emulsification Enhances the Larvicidal Potential of the Essential Oil of *Siparuna guianensis* (Laurales: Siparunaceae) *Against Aedes* (Stegomyia) aegypti (Diptera: Culicidae)	https://doi.org/10.1093/jme/tjz221	18	The authors addressed the limited water miscibility of *Siparuna guianensis* essential oil by developing a nanoemulsion to improve its biological delivery. Both the crude oil and the nanoemulsion exhibited larvicidal activity, with the nanoemulsified formulation showing superior efficacy. Phytochemical analysis identified curzerenone as the major constituent, highlighting how formulation strategies can enhance the biological performance of *S. guianensis* essential oil
Moura et al. (2020)	Antibacterial activity of *Siparuna guianensis* essential oil mediated by impairment of membrane permeability and replication of pathogenic bacteria	https://doi.org/10.1016/j.indcrop.2020.112142	31	The authors investigated the antibacterial activity of *Siparuna guianensis* essential oil against *Escherichia coli*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and *Streptococcus pyogenes*. Phytochemical analysis identified β-myrcene, epicurzerenone, and germacrenes D and B as major constituents. The oil inhibited bacterial growth without inducing cytotoxicity in human THP-1 cells and increased bacterial membrane permeability, leading to nucleotide and K^+ leakage. Molecular docking further suggested that germacrene B interacts with bacterial DNA and RNA polymerases, indicating disruption of replication as a potential mechanism of action
(Carvalho et al. 2020)	In vitro anthelmintic activity of *Siparuna guianensis* extract and essential oil against Strongyloides venezuelensis	https://doi.org/10.1017/S0022149X19000282	14	The authors assessed the anthelmintic activity of *Siparuna guianensis* essential oil and its major compound, α-bisabolol, against *Strongyloides venezuelensis* eggs and larvae using egg hatching and larval motility assays. Both the essential oil and α-bisabolol exhibited ovicidal and larvicidal effects comparable to conventional drugs and induced morphological alterations in the parasites. Notably, α-bisabolol showed efficacy similar to the crude

Table 1 continued

Authors/ Year	Article Title	DOI	Number of citations (WoS)	Brief description
				oil, indicating that this compound is largely responsible for the anthelmintic activity and highlighting *S. guianensis* essential oil as a promising alternative for strongyloidiasis treatment
Toledo et al. (2019)	Essential oil from Negramina (*Siparuna guianensis*) plants controls aphids without impairing survival and predatory abilities of non-target ladybeetles	https://doi.org/10.1016/j.envpol.2019.113153	36	The authors evaluated the insecticidal and repellent activity of *Siparuna guianensis* essential oil against Myzus persicae, as well as its selectivity toward two ladybeetle predators (*Coleomegilla maculata* and *Eriopis connexa*). The oil caused significant aphid mortality and repellency while preserving the survival and predatory performance of non-target insects. Phytochemical analysis identified β-myrcene and 2-undecanone as major constituents, and molecular docking revealed selective binding of these compounds to aphid TRP channels, suggesting a mechanistic basis for the observed insecticidal selectivity and supporting *S. guianensis* essential oil as a promising tool for pest management
P. Ferreira et al. (2019)	Prolonged mosquitocidal activity of *Siparuna guianensis* essential oil encapsulated in chitosan nanoparticles	https://doi.org/10.1371/journal.pntd.0007624	53	The authors developed chitosan nanoparticles to encapsulate *Siparuna guianensis* essential oil aiming to improve its stability and larvicidal efficacy against *Aedes aegypti*. The nanoformulation exhibited high encapsulation efficiency and promoted prolonged larvicidal activity, achieving sustained mortality for up to 19 days, with complete lethality during the first week. The formulation showed adequate selectivity toward non-target aquatic organisms (i.e., Poecilia reticulata, and Danio rerio), indicating that nanoencapsulation enhances both persistence and safety of *S. guianensis* essential oil, supporting its potential as an eco-friendly alternative for mosquito control
(Arruda et al. 2019)	Antifungal activity of plant species Negramine, *Eucalyptus* and *Jatropha* on plant pathogens	https://doi.org/10.5935/PAeT.V12.N2.04	0	The authors evaluated the antifungal activity of *Siparuna guianensis* essential oil against the plant pathogens *Rhizoctonia solani* and *Sclerotium rolfsii*. Phytochemical analysis identified myrcene as the major constituent. At low concentrations, the oil significantly reduced mycelial growth of both fungi, with complete inhibition observed at higher doses when combined with other plant oils. These findings indicate moderate antifungal activity of *S. guianensis* essential oil
Lourenço et al. (2018)	Essential oil of *Siparuna guianensis* as an alternative tool for improved lepidopteran control and resistance management practices	https://doi.org/10.1038/s41598-018-25721-0	39	The authors evaluated the insecticidal activity of *Siparuna guianensis* essential oil against the lepidopteran pests *Spodoptera frugiperda* and Anticarsia gemmatalis, including *Bacillus thuringiensis*-resistant strains. The oil exhibited high toxicity toward both species, inducing necrotic and apoptotic effects in insect cell lines while sparing human cells. Sublethal exposure also impaired reproduction, larval development, feeding, and locomotion, indicating broad physiological disruption. These findings support *S. guianensis* essential

Table 1 continued

Authors/ Year	Article Title	DOI	Number of citations (WoS)	Brief description
				oil as a promising alternative for managing Bt-resistant lepidopteran pests
Ferreira et al. (2017)	Potential use of Negramina (*Siparuna guianensis* Aubl.) essential oil to control wax moths and its selectivity in relation to honey bees	https://doi.org/10.1016/j.indcrop.2017.08.023	32	The authors evaluated the insecticidal and repellent activities of *Siparuna guianensis* essential oil against the wax moth pests Galleria mellonella and Achroia grisella, as well as its selectivity toward honey bees (*Apis mellifera*). Phytochemical analysis identified β-myrcene and 2-undecanone as the major constituents. Both moth species showed similar susceptibility to the oil, whereas honey bee foragers were 5–10 times more tolerant. The oil repelled wax moths without affecting bee foraging behavior, supporting S. *guianensis* essential oil as a selective and sustainable alternative for wax moth control
Melo et al. (2017)	Anticariogenic and Antimycobacterial Activities of the Essential Oil of *Siparuna guianensis* Aublet (Siparunaceae)	https://doi.org/10.17807/orbital.v0i0.930	17	The authors characterized the chemical composition of *Siparuna guianensis* essential oil and evaluated its antibacterial activity against oral pathogens and mycobacteria. Phytochemical analysis identified (*E,E*)-farnesol, β-myrcene, germacrene D, and siparunone as major constituents. The oil exhibited notable anticariogenic activity against *Streptococcus mutans* and moderate antimycobacterial effects against *Mycobacterium avium*, *M. tuberculosis*, and *M. kansasii*. This study represents the first evidence of anticariogenic and antimycobacterial activities of *S. guianensis* essential oil
Andrade et al. (2016)	Essential oils: in vitro activity against *Leishmania amazonensis*, cytotoxicity and chemical composition	https://doi.org/10.1186/s12906-016-1401-9	55	The authors screened essential oils from multiple plant species, including *Siparuna guianensis*, for antileishmanial activity against *Leishmania amazonensis* promastigotes and evaluated cytotoxicity in mammalian L6 cells. *S. guianensis* essential oil showed moderate inhibitory activity within the active group of tested oils. Due to this, phytochemical characterization and mechanistic analyses were prioritized for other species with superior selectivity
Andrade et al. (2015)	Biological activity of the essential oils from *Cinnamodendron dinisii* and *Siparuna guianensis*	https://doi.org/10.1590/S1517-838246120130683	34	The authors evaluated the antibacterial, antifungal, and trypanocidal activities of *Siparuna guianensis* essential oil. Phytochemical approach revealed α-pinene, β-pinene, sabinene and bicyclogermacrene as the major compounds. The oil showed low antibacterial and trypanocidal activity, but exhibited moderate antifungal effects, particularly against *Aspergillus flavus*. These findings indicate limited efficacy against bacteria and *Trypanosoma cruzi*, while supporting antifungal potential for *S. guianensis* essential oil

Table 1 continued

Authors/ Year	Article Title	DOI	Number of citations (WoS)	Brief description
Aguiar et al. (2015)	Insecticidal and Repellent Activity of *Siparuna guianensis* Aubl. (Negramina) against *Aedes aegypti* and Culex quinquefasciatus	https://doi.org/10.1371/journal.pone.0116765	59	The authors investigated the insecticidal and repellent activities of *Siparuna guianensis* essential oil against *Aedes aegypti* and *Culex quinquefasciatus* across multiple developmental stages, as well as its cytotoxic effects on *Aedes albopictus* cells. The oil exhibited high toxicity toward eggs, larvae, pupae, and adults, strong oviposition deterrence, and complete repellency at low concentrations. In cultured insect cells, the oil induced cell death likely via necrosis
Andrade et al. (2013)	Chemical Composition and Antioxidant Activity of Essential Oils from *Cinnamodendron dinisii Schwacke* and *Siparuna guianensis* Aublet	https://doi.org/10.3390/antiox2040384	68	This study reported, for the first time, the biological evaluation of *Siparuna guianensis* essential oil and identified its major constituents, including the bicyclic monoterpenes α-pinene, β-pinene, and sabinene, as well as the sesquiterpene bicyclogermacrene. However, no antioxidant activity was detected using either the DPPH radical scavenging assay or the β-carotene/linoleic acid model

polymeric microencapsulation to enhance stability, persistence, and selectivity toward non-target organisms, indicating a translational shift from exploratory bioassays to applied biotechnological development. These results suggest that while *S. guianensis* essential oil exhibits broad biological potential, its scientific consolidation has occurred primarily within the context of pest and vector management, whereas other biological properties remain comparatively underexplored.

Institutional network analysis revealed a highly centralized structure dominated by Brazilian public universities, with Universidade Federal do Tocantins (UFT) emerging as the most productive institution (n = 9 publications; 236 citations), followed by Universidade Federal de Viçosa (UFV; n = 8; 213 citations) and Universidade de Brasília (UnB; n = 5; 221 citations) (Fig. 2A-C). Universidade Federal de Lavras (UFLA) also exhibited substantial impact despite a lower number of publications (n = 5; 201 citations), indicating a high citation density per article. The institutional collaboration network (Fig. 2A) shows these universities forming a dense central hub, particularly UFT, UFV, UnB, and UFLA, while several peripheral institutions (e.g., Universidade Federal do Amazonas, Universidade Federal de Jataí, Universidade do Estado do Pará) display weak or sporadic connections. This configuration demonstrates that research on *Siparuna guianensis* essential oil is concentrated within a small group of academic centers, with limited integration of external institutions.

Interestingly, institutions associated with the academic trajectory of the most prolific author, namely UFT (current affiliation), UFV, and UnB, also rank among the most productive and cited. This convergence suggests that sustained mentorship relationships and academic mobility have contributed to the consolidation of a core research hub, facilitating continuity in experimental approaches, infrastructure sharing, and thematic focus. From a bibliometric perspective, this institutional distribution conforms to Lotka's Law, whereby a small number of institutions account for a disproportionate share of scientific output, while most contribute only sporadically. This institutional centralization likely influences the thematic profile of the field, which is characterized by repeated chemical profiling and applied bioassays, predominantly in entomological and antimicrobial contexts. The limited participation of pharmacology-oriented or clinically focused research centers may

partly explain the scarcity of mechanistic and translational studies involving mammalian systems.

Authorship analysis further reinforces this pattern (Fig. 3). A total of 174 researchers contributed to the field; however, productivity was highly asymmetric. Aguiar RWS emerged as the most prolific and influential author (8 publications; 266 citations), also serving as corresponding author in four articles. Oliveira EE followed with seven publications, 207 citations, and three corresponding authorships. The researcher collaboration network revealed a fragmented structure composed of one dominant central cluster surrounded by multiple small and weakly connected peripheral groups (Fig. 3A). Strong intragroup collaborations contrasted with limited intergroup connectivity, indicating that studies on *S. guianensis* essential oil have advanced largely through parallel efforts rather than coordinated multicenter collaborations. This modular organization favors repeated chemical characterization of similar chemotypes across independent studies, while hindering the development of unified chemometric frameworks and standardized biological protocols. Consequently, although major sesquiterpenoid constituents are consistently reported, correlations between chemical variability and biological efficacy remain poorly integrated.

A notable feature is the central role of Dr. Raimundo Wagner de Souza Aguiar, who not only leads the principal co-authorship cluster but also represents the main citation hub (Fig. 3C–E). His current affiliation (UFT) corresponds to the most productive institution identified in the analysis, while universities associated with his postgraduate training also appear among the leading contributors. This pattern highlights the formation of sustained academic lineage networks, illustrating how mentorship trajectories and institutional continuity shape thematic consolidation, methodological persistence, and infrastructure sharing. Although such dynamics enhance productivity within core centers, they may simultaneously limit methodological diversity and independent replication.

Keyword co-occurrence analysis corroborated the chemistry-centered structure of the field (Fig. 4A). The most frequent terms were "essential oil" (n = 5), followed by "natural products," "sesquiterpenes," "negramina," and "*Siparuna guianensis*" (n = 3 each), highlighting a primary focus on chemical characterization and compound identification. Among bioactivity-related keywords, insect control–associated terms such as "plant-based biorational insecticides," "repellency," and "molecular docking" appeared with moderate recurrence (n = 2), indicating an applied orientation toward vector management supported by in silico approaches (Fig. 4B). In contrast, keywords related to mammalian pharmacology or central nervous system outcomes were notably absent, while antioxidant and antitumoral themes appeared only sporadically.

According to Zipf's Law, the most frequently occurring keywords tend to represent the conceptual core of a research domain. In our present analysis, the predominance of chemically oriented terms confirms that investigations of *Siparuna guianensis* essential oil remain largely anchored in phytochemical profiling instead of mechanistic biological exploration. The limited recurrence of biological descriptors suggests that bioactivity studies are still fragmented and primarily restricted to preliminary screening models, particularly insecticidal and antimicrobial assays. This imbalance reinforces the notion the field remains centered on chemical characterization, with limited integration of pharmacological assays in mammalian models.

The selected papers were predominantly published in journals centered on natural products chemistry and phytochemistry, including *Molecules*, *Industrial Crops and Products*, *Chemistry & Biodiversity*, *Revista Brasileira de Farmacognosia*, and *Plants*. Additional publications were dispersed across journals specializing in agriculture, entomology, microbiology, environmental sciences, and complementary medicine, reflecting the interdisciplinary yet fragmented nature of the field. Despite this thematic diversity, the absence of a dominant core journal and the low recurrence of publications within individual titles indicate a scattered publication pattern, consistent with Bradford's law of bibliographic dispersion. According to this principle, emerging or weakly consolidated research areas tend to distribute their publications across a wide range of journals instead of concentrating within a limited nucleus. In the present analysis, only *Industrial Crops and Products* (n = 3) and *Molecules* (n = 2) published more than one article, while the remaining studies were scattered across 20 distinct journals (Table 2).

This dispersion suggests that research on *Siparuna guianensis* essential oil remains in an early developmental stage, lacking a consolidated disciplinary core. Most contributions are oriented toward chemical profiling and preliminary bioactivity screening, as evidenced by the predominance of chemistry-, phytochemistry-, and agriculture-focused journals.

To assess the global distribution of research on *S. guianensis* essential oil and its biological activities, the countries of corresponding authors were analyzed (Fig. 5). Scientific output was restricted to only two countries, both located in South America. Brazil accounted for the vast majority of publications (23) and citations (568), while Colombia contributed only two articles, which together received two citations (Fig. 5A-C). This pronounced geographic concentration reflects both biological and structural factors. As *S. guianensis* is native to South America (i.e., particularly Brazil), local access to plant material likely facilitates experimental investigation. Moreover, South American biomes, especially the Amazon, represent one of the world's richest reservoirs of biodiversity and bioactive natural products, encouraging regionally driven phytochemical and pharmacological research. Despite increasing globalization in science, studies on endemic species often remain geographically localized. The dominance of Brazilian research may therefore be attributed not only to species availability but also to the presence of established research groups specializing in medicinal plants and essential oils.

Chemical compositions Siparuna guianensis

The comparative analysis of the essential oils of *Siparuna guianensis* indicates that the species exhibits extensive chemical variability across different regions, plant tissues, and environmental conditions (Table 3). Based on the compiled studies, Andrade (2013, 2016), Aguiar (2015), Melo (2017), Ferreira (2017, 2019, 2020), Lourenço (2018-Gurupi 1,Gurupi 2 and Formoso do Araguaia), Arruda (2019), Santana de Oliveira et al., (2020), Moura et al. (2020, 2021), Martins RMG (2021), Costa (2022), Diniz (2022-SMA, TO, VI), Santos, et al. (2023), Barbosa et al. (2023), Montaño-Campaz (2025), and Santos et al. (2024-SG1 to SG6), it is possible to identify structured chemical patterns involving different groups of monoterpenes and sesquiterpenes, reflecting the metabolic plasticity of the species (Table 3).

The study conducted by Andrade (Andrade et al. 2013) reported a profile dominated by bicyclogermacrene (16.71%) and β-myrcene (13.14%), accompanied by relevant amounts of germacrene D (8.68%) and other sesquiterpenes. The results from Aguiar (2015) expand this perspective by demonstrating clear tissue-specific metabolism. In this study, leaves, stems (branches), and fruits were analyzed separately, resulting in distinct chemical profiles. The leaves were characterized by a marked predominance of β-myrcene (79.71%), whereas stems and fruits showed substantial amounts of germacrene D (9-9.4%) and 2-undecanone (12-26%). These patterns indicate that tissue differentiation significantly modulates terpenoid biosynthesis, reinforcing the influence of organ-specific metabolic pathways and functional roles.

The predominance of monoterpenes, especially β-myrcene, has been further supported by studies such as Melo (2017) (45.62%), Lourenço (2018-Gurupi 1,Gurupi 2, and Formoso do Araguaia, ∼ 70-81%), and Ferreira (2019) (48.59%). These findings suggest that populations from regions with warmer climates and higher irradiance, such as Tocantins, Pará, and Maranhão, tend to synthesize elevated proportions of hydrocarbon monoterpenes. Similar patterns were reported by Diniz (2022-VI: 79.47%) and (Santos et al. 2023) indicating that β-myrcene constitutes a recurrent chemical feature in several Amazonian chemotypes.

In contrast, other studies reported a dominance of hydrocarbon and oxygenated sesquiterpenes, forming a second chemical group within the species. Germacrene D appears above 10% in Aguiar (Aguiar et al. 2015), (Andrade et al. 2016), (Ferreira et al. 2017), (Arruda et al. 2019), (Santana de Oliveira et al. 2020), Ferreira (Ferreira et al. 2020), and Moura et al. (2020), with even higher values observed in Diniz (2022-TO: 25%; VI: 31.9%). These data suggest that germacrene D is associated with chemotypes enriched in sesquiterpenes, potentially influenced by environmental conditions such as clay-rich soils, shaded habitats, and elevated relative humidity.

Bicyclogermacrene, another relevant marker, was a major constituent in Andrade (Andrade et al. 2013), Aguiar (Aguiar et al. 2015), Lourenço (2018), and specific profiles described by Santos et al. (2024- SG-

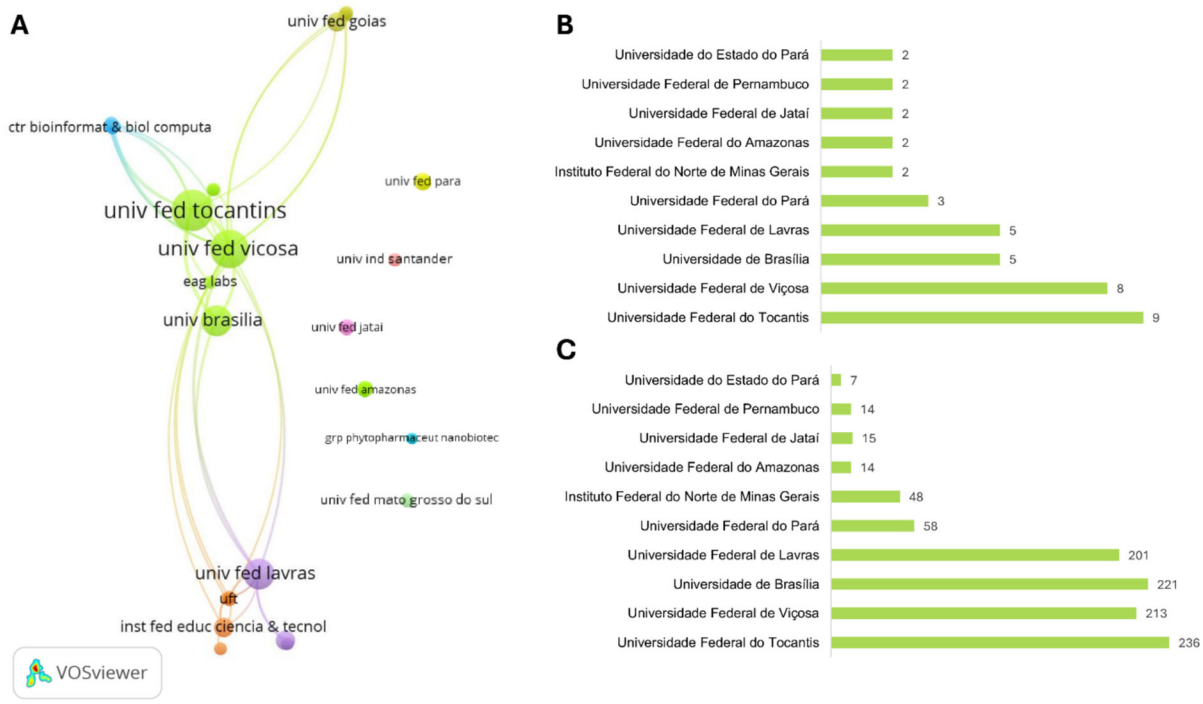


Fig. 2 Institutional contributions to the scientific literature on *Siparuna guianensis* essential oil and its biological activities: **A** institutional collaboration network, highlighting co-authorship linkages among research centers; **B** top 10 most productive institutions based on number of publications; and **C** most cited institutions according to total citation counts

5: 30.3%), demonstrating consistency among populations from different Amazonian localities. Conversely, spathulenol, typically not abundant, emerged as a dominant constituent in Barbosa et al. (2023) and in several groups examined by Santos (2024-SG1, SG4, and SG5), reaching 41.8%, 33.3%, and 43.3%, respectively. These findings indicate that some populations express a more oxidized metabolic profile, resulting in elevated levels of oxygenated sesquiterpenes.

Within this oxidized group, compounds such as curzerenone, *epi*-curzerenone, α-muurolol, and *trans*-β-elemenone also stand out, having been reported in higher concentrations in Ferreira (2020), Moura et al. (2020), Moura et al. (2021), Martins RMG (2021), and Costa (2022). These constituents are typically associated with biosynthetic pathways derived from farnesyl pyrophosphate (FPP), suggesting the activation of specific enzymatic routes under particular environmental or physiological conditions.

The data from Diniz (2022-SMA and TO) revealed markedly high levels of α-bisabolol (62.6% and 48.1%), characterizing a chemotype that differs from those reported in other Brazilian studies. Conversely, Montaño-Campaz (2025) documented a profile exclusively composed of hydrocarbon sesquiterpenes (100%), indicating a highly specialized metabolic orientation in that population. In Santos (2024-SG6), oxygenated sesquiterpenes reached 94.2%, representing another extreme within the species' chemical variability.

Collectively, the direct comparison of these studies demonstrates that *Siparuna guianensis* does not present a single predominant chemotype but instead encompasses multiple chemical groups. These include hydrocarbon monoterpene-rich chemotypes dominated by β-myrcene (as in Aguiar (2015)-leaves; Melo (2017); Lourenço (2018); Diniz (2022)-VI), hydrocarbon sesquiterpene-rich chemotypes with germacrene D and bicyclogermacrene as major constituents (Andrade (Andrade et al. 2016); Arruda (2019); (Oliveira da Silva et al. 2020)), and oxygenated sesquiterpene-rich chemotypes characterized by spathulenol, curzerenone, and α-bisabolol Moura et al. (2020); Costa (2022); Diniz (2022)-SMA; Santos (2024-SG1, SG4, SG6).

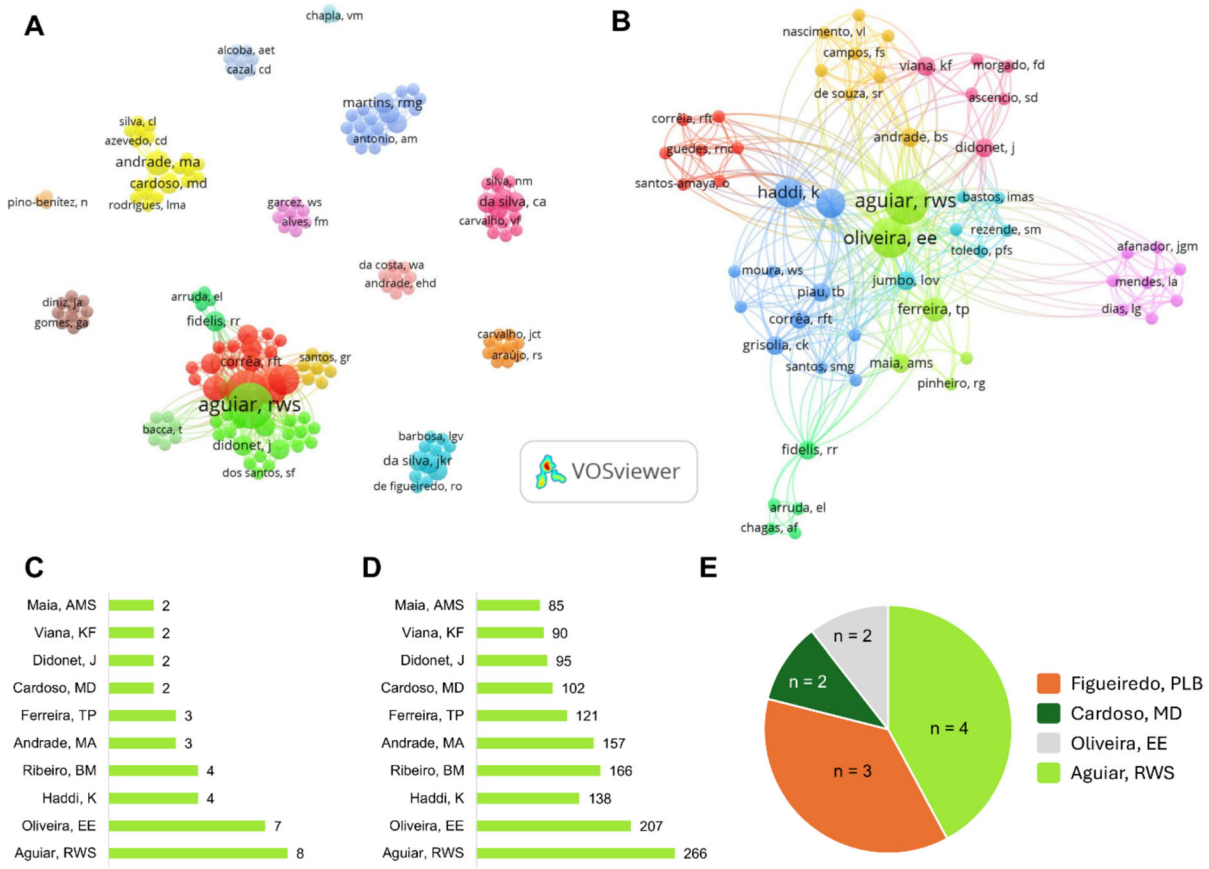


Fig. 3 Authorship landscape of *Siparuna guianensis* essential oil research: **A** global co-authorship network; **B** leading author collaboration network; **C** top authors ranked by publication count; **D** authors ranked by total citations; and **E** corresponding authorship distribution among the most prolific contributors

When considered alongside the environmental variations reported in these studies, these findings reinforce that the chemical composition of *S. guianensis* is influenced by multiple factors, including collection site, plant tissue type, plant age, microclimate, water availability, light exposure, and local ecological pressures. The coexistence of these chemotypes within the same species suggests distinct adaptive strategies, which may be related to herbivore deterrence, pathogen defense, and physiological adjustments to different environmental conditions.

Statistical analyses

The multivariate analysis based on principal component analysis (PCA) reveals a structured chemical variation across the essential oil samples, with the first two principal components (PC1 and PC2) serving as the main axes describing compositional differentiation (Fig. 6). PC1 accounts for 13.0% of the total variance (eigenvalue = 19.402), whereas PC2 explains an additional 9.2% (eigenvalue = 13.723), resulting in a cumulative variance of 22.2%. Although the explained variance may appear modest, this is expected given the high dimensionality of the dataset (149 variables) and the compositional complexity typical of essential oils, which frequently comprise numerous low-abundance constituents. Importantly, despite the distributed variance, PC1 and PC2 effectively capture the major chemical trends and reveal clear chemotypic structuring rather than random dispersion.

The distribution of samples across the PCA score space demonstrates distinct compositional groupings rather than a continuous gradient, indicating the presence of differentiated chemotypes (Fig. 6). The chemical separation along PC1 primarily reflects a

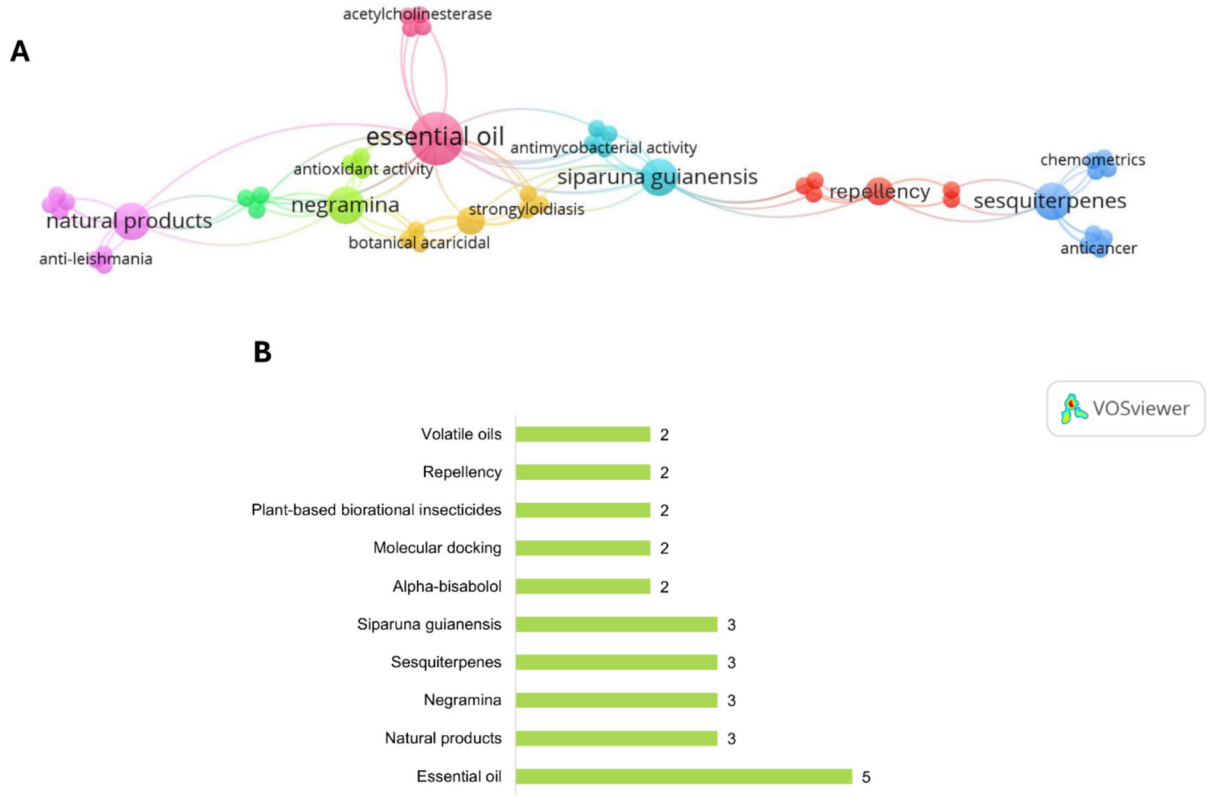


Fig. 4 Keyword co-occurrence analysis in *Siparuna guianensis* essential oil research showing **A** network visualization of co-occurring keywords and **B** the top 10 most frequent keywords across the selected publications

compositional transition from monoterpene-dominated profiles to sesquiterpene-enriched profiles. Samples positioned in the positive region of PC1 are predominantly characterized by higher relative abundances of monoterpene hydrocarbons and oxygenated monoterpenes, including β-myrcene, camphene, (E)-β-ocimene, and limonene. Conversely, samples located toward the negative side of PC1 are strongly associated with elevated proportions of sesquiterpene hydrocarbons and oxygenated sesquiterpenes. This trend demonstrates that PC1 effectively functions as a chemical axis discriminating lighter, more volatile monoterpene fractions from heavier, more structurally complex sesquiterpene components.

The loading structure associated with PC1 further reinforces this interpretation (Fig. 7). The compounds exhibiting the strongest negative contributions to PC1 include sesquiterpene hydrocarbons (α-ylangene, α-gurjunene, cis-muurola-3,5-diene, trans-cadina-1,4-diene, β-ylangene, β-selinene) and oxygenated sesquiterpenes (germacrone, trans-muurola-4(14),5-diene). These constituents collectively drive the negative polarity of PC1 and characterize a distinct sesquiterpene-rich chemotype. In contrast, positive contributions to PC1 are primarily associated with monoterpene constituents such as β-myrcene, camphene, (*E*)-β-ocimene, and limonene. Although the magnitude of these positive loadings is lower compared to the dominant sesquiterpene contributors, their combined influence clearly distinguishes samples enriched in monoterpene hydrocarbons from those dominated by sesquiterpenes.

The second principal component (PC2) introduces an additional layer of chemical differentiation (Fig. 7). PC2 is particularly influenced by variations in oxygenated compounds, with positive loadings observed for β-myrcene, 2-undecanone, and (*E*)-β-ocimene, whereas negative contributions are more closely associated with oxygenated sesquiterpenes. This distribution indicates that PC2 captures chemical variability related not only to molecular class but also to functionalization patterns, effectively

Table 2 Journals publishing articles focused on *Siparuna guianensis* essential oil and its biological activities

Journal	Number of articles	Impact factor	Primary Research Field
Plants (Basel)	1	4.1	Plant science/Phytochemistry
Boletin Latinoamericano y del Caribe de Plantas Medicinales y Aromáticas	1	0.7	Ethnopharmacology/Medicinal plants
Horticulturae	1	3.0	Horticulture/Crop science
Chemistry & Biodiversity	1	2.5	Natural products chemistry
Revista Virtual de Química	1	0.4	Analytical/Organic chemistry
Journal of the Brazilian Chemical Society	1	1.3	Chemical sciences
International Journal of Acarology	1	1.0	Entomology/Pest control
Revista Brasileira de Farmacognosia/Brazilian Journal of Pharmacognosy	1	1.6	Pharmacognosy/Natural products
Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy	1	4.6	Analytical spectroscopy/Identification of biocompounds
Industrial Crops and Products	3	6.2	Agricultural biotechnology/Plant-based products
Molecules	2	4.6	Chemistry/Bioactive compounds
Journal of Medical Entomology	1	2.0	Medical entomology/Vector control
Journal of Helminthology	1	1.6	Parasitology
Environmental Pollution	1	7.3	Environmental toxicology
PLOS Neglected Tropical Diseases	1	3.4	Tropical medicine/Infectious diseases
Applied Research & Agrotechnology	1	NA	Agricultural sciences
Scientific Reports	1	3.9	Multidisciplinary sciences
Orbital: The Electronic Journal of Chemistry	1	0.7	Chemistry
BMC Complementary and Alternative Medicine	1	NA	Complementary medicine/Pharmacology
Brazilian Journal of Microbiology	1	1.9	Microbiology
PLOS One	1	2.6	Multidisciplinary sciences
Antioxidants	1	6.6	Oxidative stress/Bioactive compounds

Legend: NA = journals without impact factor (IF)

distinguishing hydrocarbon-rich profiles from those enriched in oxygenated derivatives. Taken together, PC1 and PC2 delineate a multivariate space that integrates both biosynthetic class differentiation and degree of oxygenation.

The hierarchical cluster analysis (HCA) was conducted to further interrogate the structure revealed by PCA and to clarify the degree of compositional similarity among samples (Fig. 8). The dendrogram structure and amalgamation schedule demonstrate that the samples do not form a single continuous gradient of similarity but rather cluster into discrete compositional groupings, consistent with the existence of distinct chemotypes within the dataset. The earliest amalgamation step occurs at a similarity level exceeding 98%, indicating that at least one pair of samples shares an almost identical chemical composition. Specifically, samples R9 and R12 exhibited the highest similarity (98.36%), forming a tightly clustered pair that subsequently merged with additional samples into a highly cohesive group. This high similarity suggests that, within certain subsets of the dataset, the essential oil profiles are highly conserved.

As clustering progresses, additional samples are incorporated into clusters at progressively lower similarity thresholds, with multiple cluster mergers occurring above 90% similarity. The presence of several early-stage mergers at similarity levels exceeding 93-96% supports the presence of tightly grouped sample sets rather than diffuse compositional variation. These early clusters correspond to samples with closely aligned sesquiterpene profiles,

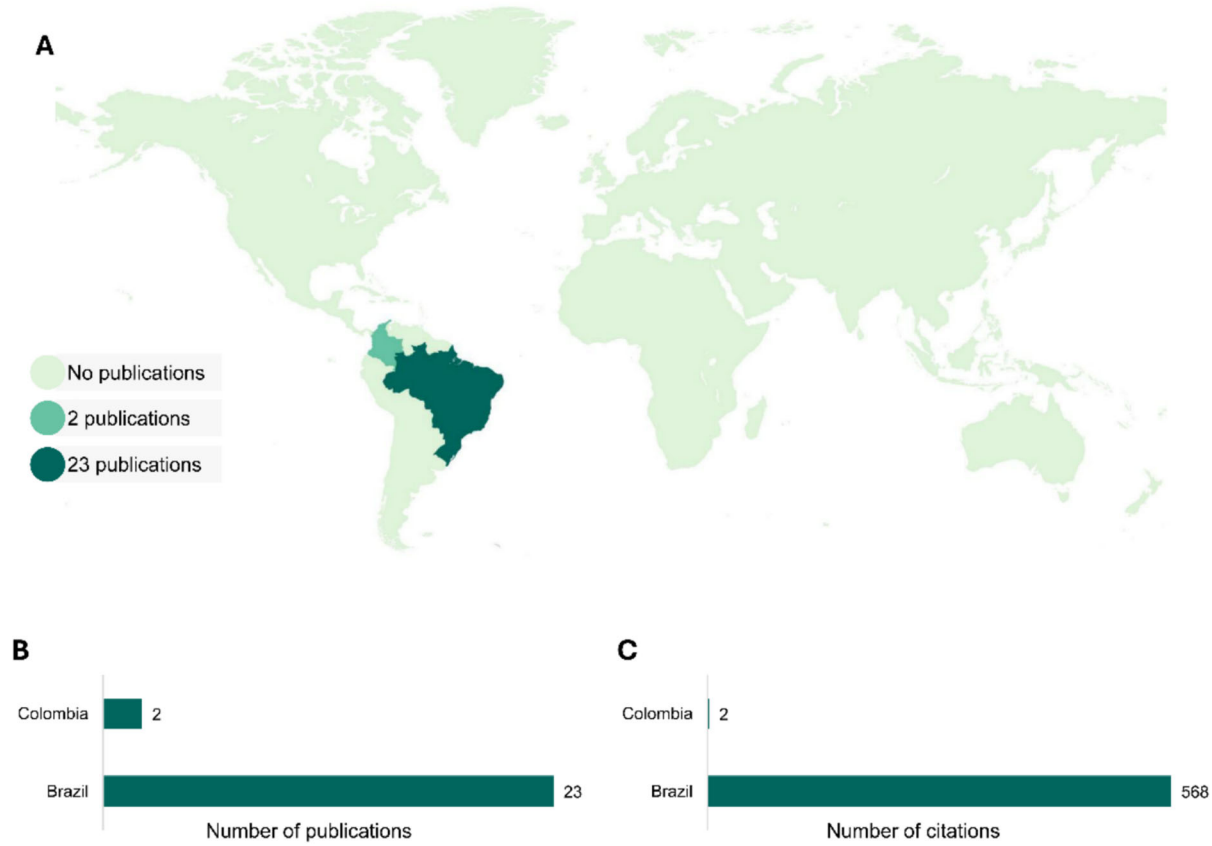


Fig. 5 Global distribution of publications on *Siparuna guianensis* essential oil and its biological activities: world map highlighting contributing countries (**A**), number of publications by country (**B**), and total citations by country (**C**)

particularly those characterized by high relative abundances of germacrone, α-ylangene, α-gurjunene, *cis*-muurola-3,5-diene, and related sesquiterpene hydrocarbons. The recurrence of these compounds in samples that cluster at high similarity levels indicates that sesquiterpene-rich compositions represent a dominant chemical signature driving group formation.

A particularly well-defined cluster is observed in Cluster 2, which comprises four samples (R2, R8, R9, and R12) and displays a low within-cluster sum of squares (1.1319) and a small average distance from the centroid (0.504). These statistics indicate a highly cohesive group with minimal internal compositional dispersion. The chemical coherence of this cluster is largely attributable to elevated and relatively stable concentrations of germacrene-type and muurolane-type sesquiterpenes, which contribute strongly to the negative loadings on PC1. The tight clustering of these samples suggests that they may represent a distinct sesquiterpene-dominated chemotype.

In contrast, Cluster 9 exhibits a markedly higher within-cluster sum of squares (17.3579) and a substantially greater average distance from the centroid (2.946). Compared to Cluster 2, Cluster 9 shows an approximately 15-fold higher within-cluster sum of squares (17.36 vs. 1.13), indicating pronounced internal heterogeneity. This cluster aggregates samples that share broad chemical similarities but differ substantially in the relative abundance of key constituents. The heterogeneity within this group appears to be driven by variable proportions of both monoterpenes and sesquiterpenes, including β-myrcene, (*E*)-β-ocimene, β-selinene, and oxygenated sesquiterpenes. The elevated dispersion within this cluster suggests that these samples may represent transitional chemical profiles or intermediate phenotypes between more clearly defined chemotypes.

Importantly, not all samples cluster at high similarity levels. Several samples are only incorporated into major clusters at similarity thresholds below 50%,

Table 3 Chemical composition of *Siparuna guianensis* (Aubl.) collected in different years and locations, compiled from independent literature studies, considering variations associated with the collection period, plant organ analyzed, and experimental

Classes	Constituents	[R1]	[R2.1]	[R2.2]	[R2.3]	[R3]	[R4]	[R5]	[R6.1]
MH	α-Thujene								
MH	α-Pinene	1.83	0.38		3.06	1.83	1.9	0.63	0.33
MH	Camphene	0.04	0.11				0.1	0.15	0.1
MH	β-Pinene	0.86	0.17				0.9	0.37	0.15
MH	β-Myrcene	13.14	79.71	26.91	16.42	13.14	16	45.62	69.3
MH	α-Phellandrene	0.03					0.1		
MH	δ-3-Carene	0.72		1.36			0.1		
MH	*p*-Cymene						0.8		
MH	Limonene	1.23	0.67	7.91	2.87		1.4	1.49	0.58
MH	β-Phellandrene	0.06					0.1		
MH	Sylvestrene								
MH	(*Z*)-β-Ocimene	0.03							
MH	(*E*)-β-Ocimene		0.64					1.07	0.57
MH	Terpinolene		0.16					1.21	0.15
MO	Ipsdienol							0.18	
O	*n*-Nonanol				5.45			0.18	
MO	α-Terpineol							0.12	
O	2-Undecanone	1.69	14.58	12.8	26.5	1.69	1.8	17.83	8.37
O	*trans*-Pinocarvyl acetate								
SH	δ-Elemene	0.58		20.92			0.5	1.48	
SH	α-Cubebene	0.04					0.1	0.08	
SH	α-Ylangene								
SH	α-Copaene	0.27	0.13				0.3	0.21	
SH	β-Bourbonene	0.31					0.2		
SH	β-Cubebene	0.18					0.2	2.62	
SH	β-Elemene	2.08	0.36	1.36			2		0.37
SH	α-Funebrene								
SH	(*Z*)-Caryophyllene			1.31					
SH	α-Gurjunene								
SH	(*E*)-Caryophyllene	1.12	0.35				1.2	0.11	0.39
SH	β-Ylangene								
SH	γ-Elemene	0.05					0.1	8.42	
SH	β-Copaene	0.04					0.1		
SH	α-Guaiene								
SH	Aromadendrene	0.04					0.2		
SH	6,9-Guaiadiene								
SH	*cis*-Muurola-3,5-diene								
SH	α-Humulene	2.07					2	0.17	
SH	Alloaromadendrene	0.05					0.1		
SH	*cis*-Cadina-1(6),4-diene								
SO	9-*epi*-(*E*)-Caryophyllene								
SH	γ-Gurjunene								
SH	γ-Muurolene							2.31	
SH	γ-Amorphene								

Table 3 continued

Classes	Constituents	[R1]	[R2.1]	[R2.2]	[R2.3]	[R3]	[R4]	[R5]	[R6.1]
SH	β-Selinene	0.2					0.3		
SH	Dauca-5,8-diene								
SH	Germacrene D	8.68	0.8	9.42		8.68	10		0.64
SH	Elixene							5.67	
SO	Shyobunone								
SH	*trans*-Muurola-4(14),5-diene								
SO	Curzerene	2.15							
SH	Bicyclogermacrene	16.71	1.21	7.79		16.71			1.76
SH	α-Muurolene	1.17					1.2	2.6	
SH	Germacrene A		0.42						0.5
SH	α-Amorphene								
SH	γ-Cadinene	2.13	0.09				2		0.1
SH	*trans*-Calamenene	0.29					0.2		
SH	β-Sesquiphellandrene								
SH	δ-Cadinene	1.04				1.04	1	0.94	
SO	*epi*-Shyobunone								
SH	Zonarene								
SH	Selina-4(14),7(11)-diene								
SH	*trans*-Cadina-1,4-diene								
SH	α-Cadinene								
SH	α-Calacorene								
SH	Selina-3,7(11)-diene								
SO	Elemol							0.21	
SH	Germacrene B	2.34	0.11				2		0.21
SO	Ledol						2.1		
SO	Spathulenol	4.16	0.25	6.09		4.16	0.3	3.87	0.21
SO	Caryophyllene oxide	0.45					0.4		
SO	allo-Hedycaryol								
SO	*cis*-β-Elemenone								
SO	Globulol	0.4					0.2	0.24	
SO	Viridiflorol	3					1.4		
SO	*trans*-β-Elemenone								
SO	Curzerenone								
SO	Humulene epoxide II	0.63							
SO	*iso*-Shyobunone								
SO	*epi*-Curzerenone								
SO	1-*epi*-Cubenol	0.15					0.5		
SO	γ-Eudesmol								
SO	*epi*-α-Cadinol		0.18				3		0.25
SO	*epi*-α-Muurolol								
SO	β-Eudesmol	1.02					1		
SO	α-Muurolol								
SO	Cubenol								
SO	τ-Muurolol								
SO	Agarospirol				3.85				

Table 3 continued

Classes	Constituents	[R1]	[R2.1]	[R2.2]	[R2.3]	[R3]	[R4]	[R5]	[R6.1]
SO	δ-Cadinol							0.67	
SO	α-Cadinol	6.09	0.12	1.42	1.05	4.14	1.8	0.28	0.19
SO	Himachalol								
SO	Atractylone								
SO	Cadin-4-en-10-ol								
SO	Siparunone						14.6		
SO	*epi*-α-Bisabolol								
SO	α-Bisabolol	3.53				3.35	2.5		
SO	Germacrone								
O	Mint sulfide								
SO	(*E*,*E*)-Farnesol						18		
O	(2*E*,6*E*)-Farnesyl acetate								
	Monoterpene hydrocarbons	17.94	81.84	36.18	22.35	14.97	21.4	50.54	71.18
	Oxygenated monoterpenoids	0	0	0	0	0	0	0.3	0
	Sesquiterpene hydrocarbons	39.39	3.47	40.8	0	26.43	23.7	24.61	3.97
	Oxygenated sesquiterpenoids	21.58	0.55	7.51	4.9	11.65	45.8	5.27	0.65
	Others	1.69	14.58	12.8	31.95	1.69	1.8	18.01	8.37
	Total identified	80.6	100.44	97.29	59.2	54.74	92.7	98.73	84.17

Classes	[R6.2]	[R6.3]	[R7]	[R8]	[R9]	[R10]	[R11]	[R12]
MH			0.94					
MH	0.38	0.51		0.4	**0.33**	4.54		
MH	0.11	0.12		0.11				
MH	0.17	0.2		0.17	0.04			
MH	79.71	75.8	48.59	74.93	0.22		**39.67**	
MH					0.15			
MH								
MH								
MH	0.67	0.74		0.66				
MH								
MH					0.51			
MH								
MH	0.64	0.47		0.56	0.07			
MH	0.16	0.14		0.15				
MO								
O								
MO								
O	9.58	10.8	5.43	9.58	0.38		6.25	
O								
SH					5.38	**2.71**		
SH					0.48	0.84		
SH					0.12			
SH	0.13	0.25		0.19	1.1	1.21		
SH					0.52			
SH					3.34	1.63		

Table 3 continued

Classes	[R6.2]	[R6.3]	[R7]	[R8]	[R9]	[R10]	[R11]	[R12]
SH	0.36	0.51		0.41	0.55	3.02		
SH					0.03			
SH								
SH					0.06			
SH	0.35	0.13		0.29	0.03			
SH					4.14	3.39		
SH			7.39		7.04	4.17	3.5	
SH						1.21		
SH					0.23			
SH					0.19	0.67		
SH					0.12			4.1
SH					1.4			
SH					0.86	0.94		
SH					0.29			
SH					0.35	0.5		
SO					0.09			
SH					0.49			
SH			0.67		0.7	7.35	1.34	
SH					0.48			
SH					1.61			
SH						0.64		
SH	0.8	1.04	9.93	0.82	7.61		14.34	
SH							3.5	
SO								**2.7**
SH					0.63	0.76		
SO					7.1	10.48	4.91	
SH	1.21	1.59		1.52				
SH					1.2	0.49		6.1
SH	0.42	0.21		0.37	0.02			
SH								
SH	0.09	0.19		0.12	0.39	5.24		
SH								
SH								
SH			0.57		1.86	3.74	1.25	2.2
SO								18.9
SH					0.48	0.34		
SH					1.04			
SH					0.45			
SH						0.77		
SH								3.6
SH					0.25			
SO						0.81		
SH	0.11	0.15	1.59	0.15	1.88	0.84	2.93	
SO								
SO	0.25	0.62	2.19	0.36		1.66	1.03	3.7

Table 3 continued

Classes	[R6.2]	[R6.3]	[R7]	[R8]	[R9]	[R10]	[R11]	[R12]
SO								
SO					0.51			
SO					1.43			
SO								
SO								
SO					11.78			
SO						18.86		
SO								
SO								23.9
SO			19.31				18.16	
SO					0.97	1.3		
SO								4.8
SO	0.18	0.32		0.25				
SO								
SO								
SO						11.75		
SO					1.15	0.93		
SO								5.1
SO								
SO								
SO	0.12	0.26	0.62	0.19		1.36	1.32	
SO								
SO					18.65			
SO								
SO								
SO								2.2
SO								
SO					5.26	2		
O					0.05			
SO								
O								2.2
	81.84	78	49.53	76.98	1.32	4.54	39.67	0
	0	0	0	0	0	0	0	0
	3.47	4.07	20.15	3.87	45.32	40.46	26.86	16
	0.55	1.2	22.12	0.8	46.94	49.15	25.42	61.3
	9.58	10.8	5.43	9.58	0.43	0	6.25	2.2
	95.44	94.1	97.23	91.23	94.01	94.15	98.2	79.5

Classes	Constituents	[R13]	[R14]	[R15.1]	[R15.2]	[R15.3]	[R16]	[R17]
O	3-methylcyclopentyl acetate						0.22	
MH	α-Thujene							
MH	α-Pinene			0.1				
MH	Camphene							
MH	β-Pinene						0.22	
MH	β-Myrcene	39.16		0.4	1.4		79.5	

Table 3 continued

Classes	Constituents	[R13]	[R14]	[R15.1]	[R15.2]	[R15.3]	[R16]	[R17]
O	3-methylcyclopentyl acetate						0.22	
MH	α-Phellandrene				0.8			
MO	2,6-Dimethyl-3,5,7-octatriene-2-ol,Z,Z-	1.14						
MH	δ-3-Carene				1.6			
MH	α-Terpinene				0.5			
MH	*p*-Cymene							
MH	Limonene	0.95			0.1		0.85	
MH	β-Phellandrene							
MH	Sylvestrene							
MH	(Z)-β-Ocimene						0.26	
MH	(*E*)-β-Ocimene						1.21	
MO	*cis*-Sabinene hydrate (IPP vs, OH)						0.9	
MH	Terpinolene				17			
MO	Geraniol						0.54	
MH	2,5,6-trimethyl-1,3,6-heptatriene						0.29	
O	2-Undecanone	6.25					8.65	
O	*trans*-Pinocarvyl acetate						0.93	
SH	Bicycloelemene			0.2				
SH	δ-Elemene	1.99				0.5		0.1
SH	α-Cubebene			1.4				0.2
O	Cyclohexanol, 2-methylene-3-(1-methylethyl)- acetate, *cis*-	2.3						
SH	Cyclosativene							0.2
SH	α-Copaene					1.6		0.7
SH	β-Bourbonene	0.76						2.1
SH	β-Cubebene							
SH	β-Elemene					4.2	0.35	1.1
SH	α-Funebrene							
SH	(Z)-Caryophyllene						0.22	
SH	β-Longipinene							
O	dihydro-α-Ionone							0.2
SH	(*E*)-Caryophyllene			0.2		1.4		
SH	β-Ylangene	1.64						0.2
SH	α-*trans*-Bergamotene						0.24	
SH	γ-Elemene					0.2		
SH	β-Copaene	9.23						0.1
SH	α-Guaiene							
SH	Aromadendrene							0.1
SH	(Z)-β-Farnesene			0.2				
SH	α-Humulene					0.5		
SH	(*E*)-β-Farnesene			1.2	0.7		0.36	
SH	Alloaromadendrene					1		
SO	9-*epi*-(*E*)-Caryophyllene							0.2
SH	γ-Gurjunene							
SH	γ-Muurolene							0.5

Table 3 continued

Classes	Constituents	[R13]	[R14]	[R15.1]	[R15.2]	[R15.3]	[R16]	[R17]
O	3-methylcyclopentyl acetate						0.22	
SH	γ-Amorphene							
SH	β-Selinene							1.4
SH	Dauca-5,8-diene							
SH	Germacrene D			0.5	1.7	25	3.12	0.2
SH	Elixene	5.88						
SO	Shyobunone		2.7					
SH	*cis*-β-Guaiene		4.1					
SH	*trans*-Muurola-4(14),5-diene							
SO	*epi*-Cubebol	1.59						1.5
SH	δ-Amorphene							
O	2-Tridecanone							
SO	Curzerene	3.91						
SH	Bicyclogermacrene			3.5	0.3	10		
SH	α-Muurolene		6.1					0.8
SH	Epizonarene							
SH	β-Bisabolene			3.3	1.5			
SH	α-Bisabolene				2.5			
SH	α-Amorphene							
SH	γ-Cadinene					6.1		
SO	Butylated hydroxytoluene						0.22	
SO	Cubebol							2.1
SH	*trans*-Calamenene							
SH	β-Sesquiphellandrene			0.7	1			
SH	δ-Cadinene		2.2			3		
SO	*epi*-Shyobunone		18.9					
SH	*cis*-Calamenene							1.6
SH	α-Calacorene							
SH	Selina-3,7(11)-diene							
SO	Elemol		3.6	5.1		4		0.5
SH	Germacrene B	3.97				0.8	1.17	
SO	*epi*-Longipinanol							
SO	Ledol					1.2		
SO	Spathulenol	2.73	3.7			1.1	0.75	12.4
SO	Caryophyllene oxide							1.8
SO	β-Copaen-4α-ol							0.3
SO	Viridiflorol							0.7
SO	Fokienol							
SO	Khusimone							
SO	Curzerenone	16.02						16.4
SO	Humulene epoxide II							
SO	β-Atlantol							
SO	*iso*-Shyobunone		23.9					
SO	*epi*-Curzerenone							
SO	1-*epi*-Cubenol							3.2

Table 3 continued

Classes	Constituents	[R13]	[R14]	[R15.1]	[R15.2]	[R15.3]	[R16]	[R17]
O	3-methylcyclopentyl acetate						0.22	
SO	γ-Eudesmol		4.8	0.7		1.8		
SO	*epi*-α-Muurolol		5.1					
SO	β-Eudesmol			0.4		4.4		
SO	2-*epi*-β-Cedren-3-one							
SO	α-Muurolol			0.6		5.6		3.1
SO	Cubenol							1.5
SO	τ-Muurolol							
SO	Agarospirol							
SO	α-Eudesmol							4.8
SO	Pogostol							1.7
SO	δ-Cadinol							
SO	α-Cadinol	0.86			3.2	9.4		
SO	Himachalol							
SO	Atractylone							
SO	Cadin-4-en-10-ol							
SO	*allo*-Himachalol							
SO	*trans*-Calamenen-10-ol							0.3
SH	Cadalene							0.5
SO	Mustakone							0.7
SO	*epi*-α-Bisabolol		2.2					
SO	α-Bisabolol			62.6	48.1			
SO	Junicedranol							
SO	Germacrone							
SO	*iso*-Bicyclogermacrenal							
SO	Eremofilone							
O	Mint sulfide							
O	Reynosin	0.72						
SO	Drimenol							13.7
SO	α-Costol		2.2					
SO	*iso*-Furanodienone	0.79						
O	n-Hexadecanol							
SO	Drimenin							0.1
O	(*E*,*E*)-Geranyllinalool							
O	Heneicosane							
O	*n*-Tetracosane							
	Monoterpene hydrocarbons	41.25	0	0.5	4.4	0	80.8	0
	Oxygenated monoterpenoids	0	0	0	0	0	1.16	0
	Sesquiterpene hydrocarbons	33.55	12.9	10.5	6.7	50.5	14.2	7.7
	Oxygenated sesquiterpenoids	22.72	62.2	7.5	1	21.9	2.14	44
	Others	0	0	0	17	0	1.75	1.5
	Total identified	97.52	75.1	18.5	29.1	72.4	100	53.2

Classes	[R18]	[R19.1]	[R19.2]	[R19.3]	[R19.4]	[R19.5]	[R19.6]
O							

Table 3 continued

Classes	[R18]	[R19.1]	[R19.2]	[R19.3]	[R19.4]	[R19.5]	[R19.6]
O							
MH				0.1			0.1
MH		0.4		0.1			0.3
MH							
MH							
MH							
MH							
MO							
MH							
MH							
MH							
MH				0.1			
MH							
MH							
MH							
MH							
MO							
MH							
MO							
MH							
O							
O							
SH							
SH		0.2	0.2	0.2			
SH		1.5	2.4	0.3			
O							
SH							
SH	4.7	3.6	2.5	0.2	2.1	0.2	0.1
SH	5.5	3.9	2.6	0.4	3.8	0.4	1.3
SH		3.3					
SH		3.9	4.8	1.1	3.4		0.4
SH							
SH							
SH			1.6				
O							
SH	4.6	2.4	2	0.4	1.5		
SH							
SH							
SH	45.8						
SH			0.2				
SH				0.3			
SH					1.1	0.8	
SH							
SH		0.3	0.4	0.1			
SH							

Table 3 continued

Classes	[R18]	[R19.1]	[R19.2]	[R19.3]	[R19.4]	[R19.5]	[R19.6]
O							
SH							
SO		0.3			1.7		
SH							
SH		0.2				0.2	0.1
SH		1	0.5	0.8		0.8	
SH			0.6				0.3
SH			0.6				
SH	31.9		0.9				
SH							
SO							
SH							
SH							
SO							
SH		1.3					
O							
SO			2.7				0.9
SH		1.4		0.4			
SH		0.2	0.6				0.1
SH				0.4			
SH							
SH							
SH			0.3			0.2	
SH		1.3	0.5	0.6			
SO							
SO		1.3	2.7				0.9
SH							
SH							
SH	7.5	3.2	2.9	0.3	5		1.1
SO							
SH							
SH		0.1	0.5				
SH							
SO		0.5	4.4	30.3	11.5		
SH				7		1.6	
SO		2.1				36.1	
SO						0.3	
SO		41.8	13.9	9.8	33.3	43.3	11.5
SO		2.8	3.4		8.7	0.9	0.4
SO							
SO			1.8				
SO		4.6				4.4	
SO			0.2			1.4	
SO			9.6				23.9
SO							

Table 3 continued

Classes	[R18]	[R19.1]	[R19.2]	[R19.3]	[R19.4]	[R19.5]	[R19.6]
O							
SO							0.3
SO							
SO							
SO		0.4	1.5		1	0.6	
SO							
SO							
SO				4.3			
SO		4.7			2.1		
SO		2.4	0.3				2.5
SO		1.2	3.1		2.4		0.8
SO			0.4				
SO							3.7
SO				5.4			
SO							4.4
SO							
SO							
SO			2.6				
SO							
SO			1				
SO						0.3	
SO							
SH							
SO		0.3	0.7	16.5	1.2		
SO			0.3				
SO							
SO			0.4				0.1
SO							
SO			0.1			1.3	
SO					6.1	5.6	15.7
O							
O							
SO		6.4	4.3				
SO			0.2				
SO							
O			0.2				7
SO							
O			0.5				
O			0.3		0.9		
O			5.6				
	0	0.4	0	0.3	0	0	0.4
	0	0	0	0	0	0	0
	92.5	24.6	22.5	4.9	13.6	2.2	2.8
	7.5	64.1	42.5	52	61.6	89	41
	0	1.2	3.1	0	2.4	0	4.5

Table 3 continued

Classes O	[R18]	[R19.1]	[R19.2]	[R19.3]	[R19.4]	[R19.5]	[R19.6]
	100	90.3	68.1	57.2	77.6	91.2	48.7

The data presented in the table were compiled from independent studies: Andrade (2013) [R1]; Aguiar (2015), with evaluation of different plant organs (leaf, stem and fruit) [R2.1-R2.3]; Andrade (2016) [R3]; Melo (2017) [R4]; Ferreira (2017) [R5]; Lourenço (2018), with studies carried out in Gurupi 1, Gurupi 2 and Formoso do Araguaia [R6.1, R6.2, and R6.3]; Ferreira (2019) [R7]; Arruda (2019) [R8]; Santana de Oliveira et al., (2020) [R9]; Ferreira (2020) [R10]; Moura et al. (2020) [R11]; Martins and RMG (2021) [R12]; Moura et al.(2021) [R13]; Costa (2022) [R14]; Diniz (2022), with analyses performed in SMA, TO and VI [R15.1-R15.3]; Santos et al. (2023) [R16]; Barbosa et al. (2023) [R17]; Montaño-Campaz (2025) [R18]; and Santos et al. (2024), with six experimental areas (SG-1 to SG-6) [R19.1–R19.6]

indicating substantial chemical divergence relative to the core groups. These late-joining samples are characterized by either reduced sesquiterpene content or elevated levels of monoterpenes and oxygenated constituents, consistent with their separation along the positive side of PC1 in the PCA. This pattern indicates that chemical differentiation is not uniform across the dataset and that certain samples deviate significantly from the dominant sesquiterpene-rich chemotype.

The overall structure of the dendrogram reveals at least two major chemical lineages: one defined by sesquiterpene-rich compositions dominated by germacrane-, ylangane-, and muurolane-type skeletons (exemplified by Cluster 2), and another characterized by higher proportions of monoterpenes and oxygenated compounds. The alignment of these clusters with the PCA loading structure reinforces the interpretation that these groupings reflect meaningful biochemical differentiation rather than statistical artifact.

Collectively, the HCA results indicate that while a subset of samples forms highly cohesive clusters with strong chemical similarity (notably R2, R8, R9, and R12 in Cluster 2), other samples display marked divergence, resulting in a structured but heterogeneous compositional landscape. The dominant role of sesquiterpenes in defining cluster boundaries highlights the importance of these compounds as chemotaxonomic markers within the dataset, while the variable distribution of monoterpenes and oxygenated constituents contributes to finer-scale differentiation among samples.

The convergence of PCA and HCA results provides strong evidence for distinct chemotypic organization within the dataset. The primary axis of variation captured by PC1, defined by the balance between sesquiterpene-dominated and monoterpene-dominated profiles, aligns closely with the cluster groupings observed in the dendrogram. This concordance indicates that the multivariate structure is chemically meaningful and reflects underlying compositional gradients rather than statistical artifact. Collectively, these results demonstrate that the essential oil profiles exhibit discrete compositional structuring, supporting the identification of distinct chemotypes and reinforcing the value of multivariate approaches for resolving complex phytochemical datasets.

Antioxidant potential

Due to the high chemical variability of EOs, it is common to evaluate their specific antioxidant activities, which, even within the same species, may vary depending on their major chemical constituents; moreover, the different methods used to assess antioxidant activity in biological systems are relatively fast, simple, and inexpensive, which justifies their application in phytochemical studies (Huang et al. 2025). Among the selected studies, all were conducted in Brazil, and the most commonly used method was the quantitative 2,2-diphenyl-1-picrylhydrazyl (DPPH) radical scavenging assay, one of the most widely applied free radical scavenging assays according to the literature; in addition, the second most frequently used method was the lipid peroxidation assay based on the β-carotene/linoleic acid co-oxidation method (Mendonça et al. 2022).

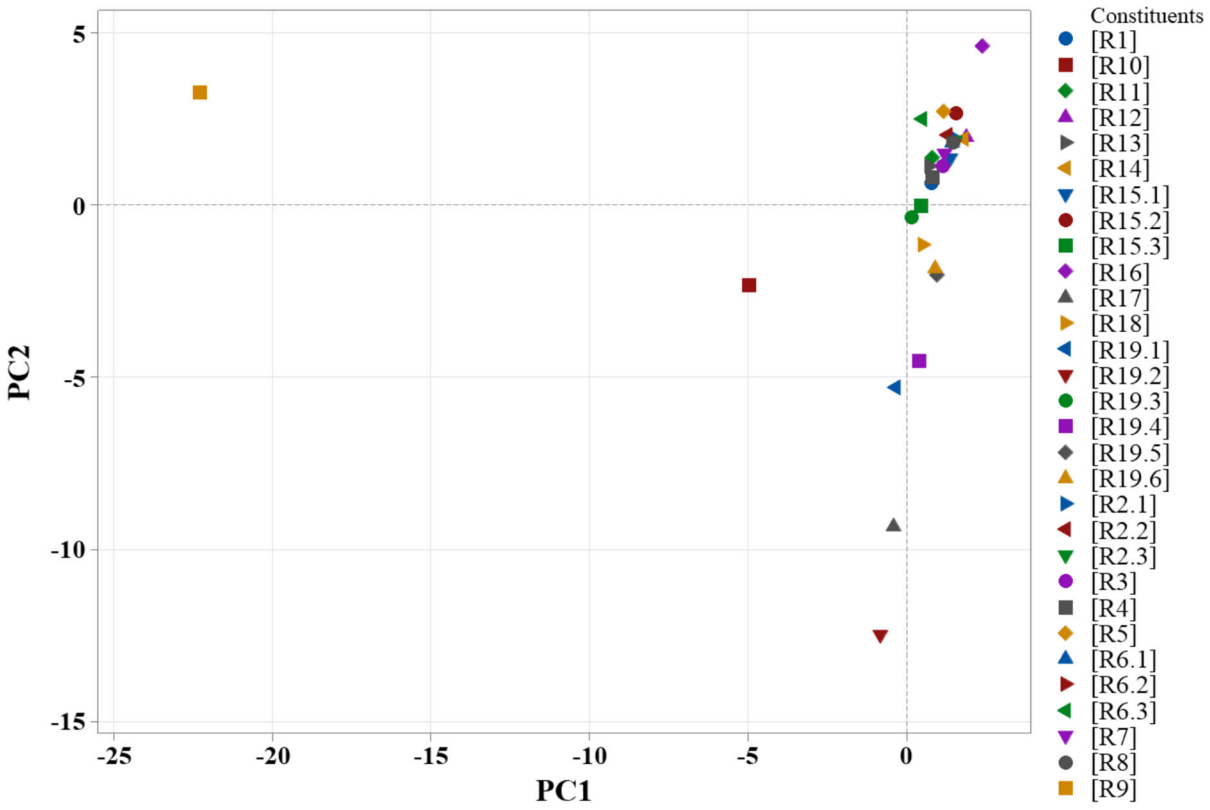


Fig. 6 PCA score plot (PC1 × PC2) showing the distribution and grouping pattern of essential oil samples from different specimens of *Siparuna guianensis* based on chemical composition compiled from literature sources. Each point represents one reference dataset, demonstrating similarity patterns and chemotypic clustering among samples. The dataset includes independent studies: Andrade (2013) [R1]; Aguiar (2015), evaluating different plant organs (leaf, stem and fruit) [R2.1-R2.3]; Andrade (2016) [R3]; Melo (2017) [R4]; Ferreira (2017) [R5]; Lourenço (2018), with collections from Gurupi 1,Gurupi 2 and *Formoso do Araguaia* [R6.1, R6.2, R6.3] [R6.1, R6.2]; Ferreira (2019) [R7]; Arruda (2019) [R8]; Santana de Oliveira et al., (2020) [R9]; Ferreira (2020) [R10]; Moura et al.(2020) [R11]; Martins and RMG (2021) [R12]; Moura et al. (2021) [R13]; Costa (2022) [R14]; Diniz (2022), with analyses from SMA, TO and VI [R15.1-R15.3]; Santos et al. (2023) [R16]; Barbosa et al., (2023) [R17]; Montaño-Campaz (2025) [R18]; and Santos et al. (2024), with six experimental areas (SG-1 to SG-6) [R19.1-R19.6]., with six experimental areas (SG-1 to SG-6) [R19.1-R19.6]. Closer points indicate higher compositional similarity among reported samples

In the study by Andrade et al. (Andrade et al. 2013), the essential oil of *S. guianensis* from the state of Minas Gerais exhibited low antioxidant activity when evaluated using the β-carotene/linoleic acid method, as it did not show IC_{50} values (concentration required to inhibit 50% of free radicals) within the tested range (> 300 μg mL^{-1}); whereas, using the DPPH method, the EO did not exhibit antioxidant activity at the tested concentrations based on the reference substances employed (BHT; thymol; and ascorbic acid).

The authors attributed this lack of antioxidant efficacy to the major composition of the EO not being compatible with the methodologies employed, as it consisted mainly of hydrocarbon mono- and sesquiterpenes, such as acyclic monoterpenes, β-myrcene, germacrene D, and bicyclogermacrene; these compounds are unable to donate hydrogen atoms to reduce the DPPH radical and exhibit low solubility in the reaction medium, since these methods use methanol or ethanol as solvents.

Similarly, Santos et al. (2024) performed the same DPPH and β-carotene/linoleic acid assays on the EO of six *S. guianensis* specimens from the state of Pará, and no significant antioxidant activity was observed in either test, with results approximately eight times lower than the Trolox standard (DPPH: $< 108.1 \pm 11.5$ TEAC; β-carotene/linoleic acid: 2.1-9.1%). The chemical compositions of the EOs were

predominantly composed of oxygenated sesquiterpenes (53.4-94.1%) and hydrocarbons (4.1-27.9%), with a marked presence of spathulenol in all specimens (mean of 25.6 ± 15.6%). Sesquiterpenes exhibit low antioxidant activity; therefore, the studied EOs did not show favorable results.

In contrast, in the study by (Santos et al. 2023), a qualitative DPPH reduction assay by thin-layer chromatography was performed, and the EO exhibited antioxidant activity according to the methodology employed, with the retention factors (RF) of the spots on the plate calculated (0.69; 0.31; and 0.10). This method uses pre-coated silica gel plates containing the EO, eluted with hexane and sprayed with a 0.2% methanolic DPPH solution, followed by exposure to artificial light; subsequently, antioxidant potential is visualized by the presence of yellowish spots, due to DPPH reduction, against the purple background. In this study, the authors used the EO from younger leaves of adult *S. guianensis* plants at the flowering stage collected in the state of Tocantins, obtaining an EO rich in monoterpenes (84.67%), with β-myrcene (79.47%) as the major compound.

Similarly to the other studies, Barbosa et al. (2023) evaluated the in vitro antioxidant potential of the EO of *S. guianensis* (DPPH and β-carotene/linoleic acid), also obtained from the state of Pará, in addition to assessing its effects on oxidative stress in animal

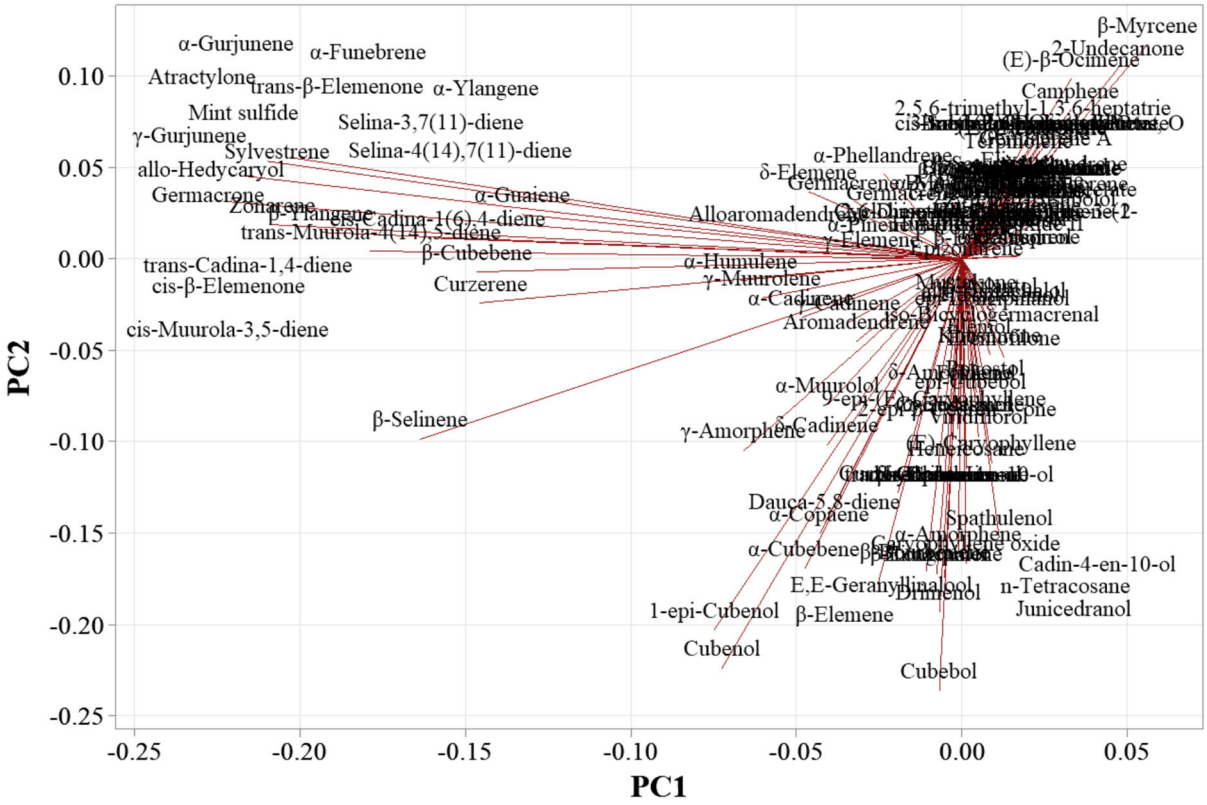


Fig. 7 PCA loading plot showing the contribution and correlation of identified chemical constituents to the principal components (PC1 and PC2) for the chemical composition of essential oils from different specimens of *Siparuna guianensis* based on compiled literature data. Each vector represents an individual compound, and its direction and magnitude indicate its influence on the variance explained by the components; variables farther from the origin contribute more strongly to sample discrimination. The dataset was compiled from independent studies: Andrade (2013) [R1]; Aguiar (2015), evaluating different plant organs (leaf, stem and fruit) [R2.1-R2.3]; Andrade (2016) [R3]; Melo (2017) [R4]; Ferreira (2017) [R5]; Lourenço (2018), with collections from Gurupi 1,Gurupi 2, and Formoso do Araguaia [R6.1, R6.2, R6.3] [R6.1, R6.2]; Ferreira (2019) [R7]; Arruda (2019) [R8]; Santana de Oliveira et al., (2020) [R9]; Ferreira (2020) [R10]; Moura et al. (2020) [R11]; Martins and RMG (2021) [R12]; Moura et al.(2021) [R13]; Costa (2022) [R14]; Diniz (2022), with analyses from SMA, TO and VI [R15.1-R15.3]; Santos et al. (2023) [R16]; Barbosa et al. (2023) [R17]; Montaño-Campaz (2025) [R18]; and Santos et al. (2024), with six experimental areas (SG-1 to SG-6) [R19.1-R19.6]

models bearing Ehrlich tumors through hepatic levels of malondialdehyde (MDA), catalase (CAT), and superoxide dismutase (SOD). Although the DPPH assay demonstrated low antioxidant activity (≈11% inhibition), approximately ten times lower than Trolox, this limitation was discussed in light of the lipophilic nature of the major sesquiterpenes, which is associated with the phytochemical profile of the oil, dominated by oxygenated sesquiterpenes and hydrocarbons. On the other hand, the lipid peroxidation assay revealed moderate antioxidant activity (≈15.5%), which was superior to that determined by the DPPH method.

Regarding the enzymatic markers, treatment with the EO in rats reduced CAT levels, indicating a loss of protection against DNA damage and damage caused by reactive species, which are functions of this enzyme. However, when compared with tumor-bearing groups, the treatment also reduced MDA levels, suggesting a decrease in the production of reactive oxygen species induced by this tumor-promoting agent, and increased SOD levels, indicating an improvement in the activity of this antioxidant enzyme, thereby demonstrating the free radical scavenging ability and the antioxidant potential of the EO of *S. guianensis* in this type of animal model.

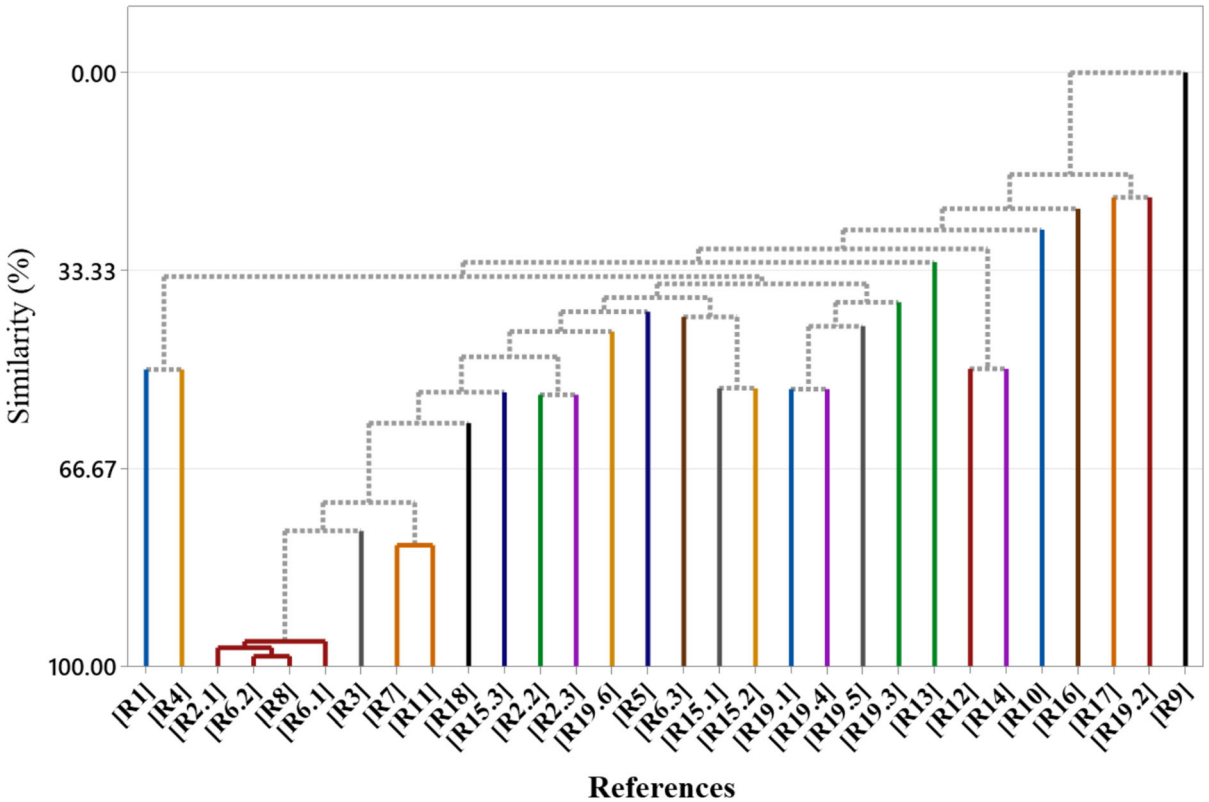


Fig. 8 Hierarchical cluster dendrogram of the chemical composition of essential oils from different specimens of *Siparuna guianensis* based on literature data. The similarity analysis was constructed from compiled compositional profiles reported in independent studies: Andrade (2013) [R1]; Aguiar (2015), evaluating different plant organs (leaf, stem, and fruit) [R2.1-R2.3]; Andrade (2016) [R3]; Melo (2017) [R4]; Ferreira (2017) [R5]; Lourenço (2018), with collections from Gurupi 1, Gurupi 2, and Formoso do Araguaia [R6.1, R6.2, R6.3]; Ferreira (2019) [R7]; Arruda (2019) [R8]; Santana de Oliveira et al., (2020) [R9]; Ferreira (2020) [R10]; Moura et al. (2020) [R11]; Martins and RMG (2021) [R12]; Moura et al. (2021) [R13]; Costa (2022) [R14]; Diniz (2022), with analyses from SMA, TO, and VI [R15.1-R15.3]; Santos et al. (2023) [R16]; Barbosa et al. (2023) [R17]; Montaño-Campaz (2025) [R18]; and Santos et al. (2024), covering six experimental areas (SG-1 to SG-6) [R19.1-R19.6]. Similarity values are expressed as percentages,standardized variables, euclidean distance, complete linkage, indicating greater compositional resemblance among specimens

biological activity potential

Repellent and insecticidal activity

The essential oil of *Siparuna guianensis* has attracted considerable pharmacological interest; however, an important aspect to be highlighted in this review is its potent activity against insect vectors and agricultural pests (Ferreira et al. 2019). Among the biological properties most consistently reported, insecticidal and repellent activities are the most frequently investigated, reflecting the growing body of research focused on this field. This increasing interest is supported by experimental evidence demonstrating the effectiveness of this species against agricultural pests as well as mosquitoes of medical importance, underscoring its potential for further exploration in vector control and pest management strategies (Aguiar et al. 2015; Ferreira et al. 2017; Ferreira et al. 2019; Montaño-Campaz et al. 2025).

An important study highlighted in this section investigated the potential of nanoemulsification of *S. guianensis* essential oil against mosquito vectors of dengue, Zika, and chikungunya viruses. The results showed that this formulation strategy substantially enhanced larvicidal activity compared to the non-formulated oil, demonstrating that appropriate delivery systems can significantly improve bioefficacy (Ferreira et al. 2019). In line with these findings, encapsulation of *S. guianensis* essential oil in chitosan nanoparticles prolonged its residual mosquitocidal activity against *Aedes aegypti* and maintained larvicidal effectiveness over time, indicating that nanostructured carriers can modulate release kinetics and protect volatile phytochemicals from rapid degradation (Ferreira et al. 2020). Additional investigations have also demonstrated insecticidal and repellent activity of the oil against *Culex quinquefasciatus*, another urban mosquito vector of epidemiological relevance, reinforcing the broad applicability of this species in mosquito control strategies (Aguiar et al. 2015).

In addition to its notable activity against mosquito larvae, the insecticidal spectrum of *S. guianensis* essential oil extends to insects of major agricultural and economic importance. A study evaluating this oil demonstrated activity against wax moth species (*Galleria mellonella* and *Achroia grisella*), pests commonly associated with beekeeping, while also showing selective toxicity with reduced adverse effects on honey bees (*Apis mellifera*), a crucial non-target pollinator (Ferreira et al. 2017). This broader spectrum of activity highlights the versatility of *S. guianensis* essential oil within integrated pest management frameworks. Another reported activity involves beetles such as *Tribolium castaneum*, an important pest of stored grains. The study indicated that essential oils from Neotropical plant species, including *S. guianensis*, may be applicable for the control of this type of agricultural pest (Pino-Benitez et al. 2024), highlighting the relevance of research exploring the potential of essential oils as botanical insecticides.

In addition to investigating direct toxicity, some studies have explored the repellent capacity of the essential oil associated with the reduction of host-seeking and oviposition behavior in insects, particularly mosquitoes. This deterrent effect is associated with the volatile terpenoid profile characteristic of the genus, rich in monoterpenes and sesquiterpenes that interfere with the olfactory and neurobehavioral pathways of insects (Aguiar et al. 2015; Montaño-Campaz et al. 2025).

Beyond direct toxicity, the repellent properties of *S. guianensis* essential oil have been implicated in reducing insect host-seeking and oviposition behaviors, particularly in mosquito species. Experimental studies evaluating both insecticidal and behavioral endpoints support a deterrent effect associated with the volatile terpenoid profile characteristic of the genus, which is rich in monoterpenes and sesquiterpenes known to interfere with insect olfactory and neurobehavioral pathways (Aguiar et al. 2015; Montaño-Campaz et al. 2025).

Importantly, the integration of formulation technologies such as nanoemulsions, biopolymeric nanoparticles, and starch-based microparticles has been shown to enhance the insecticidal performance of *S. guianensis* essential oil while potentially improving its environmental compatibility. Encapsulation approaches using biodegradable matrices can increase stability, modulate release, and prolong biological activity, helping to overcome common limitations of essential oils, such as volatility and rapid degradation under field conditions. Overall, the available experimental evidence indicates that the essential oil of *S. guianensis* represents a promising bioactive resource with insecticidal and repellent potential against urban vectors, agricultural

pests, and storage insects, supporting further research that integrates phytochemical characterization, advanced delivery systems, ecological safety, and resistance-management perspectives toward the development of sustainable pest and vector control strategies.

Antimicrobial activity

Essential oils are volatile compounds that are primarily synthesized as secondary metabolites of glucose metabolism in plants and are mostly obtained through processes such as hydrodistillation or steam distillation from various plant structures. The essential oil of *S. guianensis* contains sesquiterpenes, oxygenated sesquiterpenes, and monoterpene hydrocarbons, which predominantly exhibit hydrophobic characteristics (Santana de Oliveira et al. 2020). Gram-negative bacteria possess an outer membrane rich in lipopolysaccharides (LPS), which confers a hydrophilic surface; therefore, hydrophobic compounds are generally unable to cross these intact membranes (Cox et al. 2001).

A study conducted by Andrade et al. (Andrade et al. 2015) demonstrated that *S. guianensis* essential oil exhibited inhibitory effects against Gram-positive bacteria such as *Listeria monocytogenes* and *Staphylococcus aureus* but showed no inhibitory activity against Gram-negative bacteria. The compound α-pinene showed the highest biological activity against *L. monocytogenes*. In addition, *S. guianensis* essential oil demonstrated strong antifungal activity, with lower inhibitory concentrations against *Candida albicans*, *Cryptococcus neoformans*, *Trichophyton rubrum*, and *Aspergillus fumigatus*.

Santana de Oliveira et al., (2020) investigated the essential oil of *S. guianensis* and identified trans-β-elemenone and atractylone as the most abundant compounds in the sample. Molecular docking studies revealed positive interactions with target proteins, such as *Candida albicans* N-myristoyltransferase and *Escherichia coli* enoyl reductase. The predominance of hydrophobic interactions suggests that the ligand exhibits good complementarity with the active sites of the enzymes studied, which may contribute to increased complex stability and potential biological activity. In this context, the low RMSD values, combined with favorable energetic contributions obtained by the MM/GBSA method, reinforce the stability of the formed complexes and suggest good potential of the ligand as an inhibitor of the evaluated target proteins.

The essential oil of *S. guianensis* also demonstrated inhibitory activity against *Rhizoctonia solani*, as well as inhibition of mycelial growth of *Sclerotium rolfsii*. In this study, the major components identified were the monoterpene β-myrcene and 2-undecanone (Arruda et al. 2019).

De Jesus et al. (de Jesus et al. 2020) conducted a study to evaluate the use of essential oils as a therapeutic alternative for foodborne diseases. Compared to other oils analyzed, *S. guianensis* did not demonstrate antimicrobial activity against *E. coli* strains; however, it exhibited activity against *Staphylococcus* spp., including a multidrug-resistant strain, although this was not the most pronounced result of the study.

In a study by Melo et al. (Melo et al. 2017), the most abundant compounds identified were the acyclic monoterpene β-myrcene and the sesquiterpenes germacrene D and (*E*, *E*)-farnesol. The essential oil showed antibacterial efficacy against *Streptococcus mutans* strains. The same study also evaluated anticariogenic activity, which was evidenced by activity against *S. sobrinus* and *S. mutans*, in addition to antimycobacterial activity against *Mycobacterium tuberculosis*, *M. kansasii*, and *M. avium*. These effects were directly associated with the presence of viridiflorol compounds in the essential oil (Melo et al. 2017).

Moura et al. (2020) investigated the essential oil of *S. guianensis*, whose composition was mainly characterized by β-myrcene, germacrene D, and espicurzerenone. Antimicrobial activity was observed against *E. coli*, *Pseudomonas aeruginosa*, *Streptococcus pyogenes*, and *Staphylococcus aureus*. Even at low concentrations, inhibition of bacterial growth was observed, while higher concentrations completely inhibited growth. Additionally, the essential oil increased cellular potassium efflux. Nucleotide leakage was also observed following exposure to the essential oil, and the authors demonstrated that none of the concentrations tested caused toxicity to human monocytic cells (TPH1). The results obtained in this study indicate the potential efficacy of the essential oil of Siparuna guianensis, particularly when compared with data reported for essential oils from other species or genera (da Costa et al. 2024; Ferreira et al. 2025). When compared to a reference antibiotic, the essential

oil exhibited a greater ability to alter bacterial membrane permeability, as evidenced by increased K^+ efflux. Molecular docking analyses suggest that compounds present in the essential oil, particularly germacrene B, exhibit high affinity for active regions of bacterial DNA and RNA polymerases, potentially inducing conformational changes capable of inhibiting replication and transcription processes (Robinson and Oijen 2013).

The bacteriostatic or bactericidal effect of *S. guianensis* essential oil may be explained by the inhibition of microbial growth due to damage to membrane integrity caused by lipophilic components present in the oil, leading to loss of cellular pH homeostasis and inorganic ions (Melo et al. 2017; Santana de Oliveira et al. 2020). Furthermore, increased potassium ion efflux and nucleotide leakage suggest that the essential oil compromises the integrity of the bacterial cytoplasmic membrane, allowing the release of essential intracellular components (Roosild et al. 2010; Bajpai et al. 2013). Additionally, it has been demonstrated that the phytochemical composition of *S. guianensis* essential oil is dependent on seasonality and collection site, resulting in variation of major and minor compounds. Overall, the findings indicate that Gram-positive bacteria are more sensitive to the effects of the essential oil, likely due to the volatile compounds present in its composition.

Antiparasitic activity

Few studies have demonstrated the activity of *S. guianensis* against intracellular protozoa. In an in vitro analysis, the ethanolic extract, ethyl acetate fraction, aqueous fraction, volatile oil, and isolated α-bisabolol from *S. guianensis* Aubl. showed activity against the RH2F1 strain of *Toxoplasma gondii* in fibroblasts (Souza et al. 2021). After parasite infection, the fractions exhibited partial modulation of intracellular proliferation. When used as a pretreatment, the ethanolic extract and the ethyl acetate fraction directly inhibited the parasite (Souza et al. 2021).

The essential oils obtained from the leaves of different chemotypes of *S. guianensis*, rich in α-bisabolol and germacrene D, demonstrated acaricidal activity against *Rhipicephalus microplus*. In larval proliferation assays, samples containing α-bisabolol exhibited a 100% mortality rate, although all essential oils caused high mortality in *R. microplus*. In addition, α-bisabolol–rich samples showed a reduction in both the number and viability of eggs (Diniz et al. 2022).

Anthelmintic properties were evaluated for the ethanolic extract, essential oil, aqueous and ethyl acetate fractions of *S. guianensis*, as well as for the compound α-bisabolol, against eggs of *Strongyloides venezuelensis*. In larval motility and egg hatching assays, all samples exhibited ovicidal activity comparable to albendazole, in addition to inducing morphological alterations in the eggs (Carvalho et al. 2020).

In a study evaluating sixteen essential oils from *S. guianensis*, activity was observed against the promastigote forms of *Leishmania (L.) amazonensis* (Andrade et al. 2016). The essential oil from *S. guianensis* leaves, composed of β-myrcene, germacrene D, and bicyclogermacrene, was the most effective in inhibiting protozoan growth; however, it exhibited higher cytotoxicity toward L6 and L cells compared with the other samples (Andrade et al. 2016). Overall, extracts, essential oils, and fractions of *S. guianensis* display inhibitory activity against parasites such as ticks, helminths, and protozoa, indicating potential for the treatment of related diseases.

Antitumoral activity

Traditionally known in the Amazon region as "capitiú," the species *S. guianensis* exhibits high variability in the chemotypes of its essential oils. Recent chemical studies have identified curzerenone (16.4%), drimenol (13.7%), and spathulenol (12.4%) as major constituents of the essential oil of this plant collected in Brazil.

Curzerenone has been associated in bioactivity studies with potential cytotoxic effects in tumor cell models, including the ability to induce apoptosis in cancer cell lines, suggesting a possible antitumor role for its derivatives. Furthermore, spathulenol is frequently linked to immune system modulation and demonstrates antiproliferative properties in different biological systems, reinforcing the pharmacological interest in oxygenated sesquiterpenes present in the essential oil of *S. guianensis.* According to Barbosa et al. (2023), this profile had not been previously described in the literature, strengthening the influence of edaphoclimatic, seasonal, and geographic factors on the synthesis of secondary metabolites. This finding contributes to expanding the phytochemical knowledge of the species and highlights the importance of

regional studies with Amazonian medicinal species (Martins et al. 2021).

Barbosa et al. (2023) conducted a study that integrates chemical analyses (GFID and GC-MS), in vitro antioxidant assays (DPPH and β-carotene/linoleic acid) and in vivo evaluation based on the experimental oncology model with the Ehrlich ascites carcinoma (EAC) model to test the efficacy of S. guianensis essential oil. In this context, in vivo tests showed that after administration of 100 mg/kg of the oil, it not only reduced the volume of ascitic fluid but also preserved the viability of hepatocytes, suggesting low systemic toxicity compared to conventional chemotherapeutic agents. Decreases in malondialdehyde (MDA) levels and increased superoxide dismutase (SOD) activity in liver tissues indicated a protective effect against tumor-induced oxidative stress. These results are consistent with the literature that associates the control of oxidative stress with the inhibition of tumor progression (Marnett 1999).

The antitumor effect of S. guianensis essential oil can be defined by its complex composition of sesquiterpenes, which has a synergistic effect that inhibits and protects the body against oxidative damage.

In silico analyses

In silico studies on *S. guianensis* have provided significant insights into the potential molecular mechanisms underlying experimentally observed biological activities. These computational approaches have been primarily applied to establish correlations between the major sesquiterpenes present in the essential oil and relevant biological targets, employing methodologies such as homology modeling, molecular docking, molecular dynamics simulations, and binding energy calculations. Such strategies enable the rationalization of bioactivity at the molecular level and support the identification of key ligand–target interactions (Toledo et al. 2019; Santana de Oliveira et al. 2020; Martins et al. 2021; Costa et al. 2022; Montaño-Campaz et al. 2025).

Research by Toledo et al. (2019) regarding insecticides and selectivity offered the use of computational modeling specifically with β-myrcene and 2-undecanone, hypothesizing that the main constituents of the essential oil of *S. guianensis* act as TRP (Transient Receptor Potential) channels of *Myzus persicae*. Therefore, homology modeling (Swiss-Model) integrated with molecular docking (Autodock 4) was employed in the study. The main results highlight that both compounds bound to the TRP channel, specifically the lipid activation site where 2-undecanone formed an immunity bond with Ser450 and hydrophobic van der Waals interactions with residues Y470, L451, M467, D446, S448, and T449, showing a modulating ligand profile. β-Myrcene exhibited exclusively hydrophobic interactions with alkyl/π-alkyl type contacts with Y470, L451, M467, high compatibility with the lipid environment of the channel, which is consistent with the nonpolar nature of the compound (Adams 2007). Finally, when evaluating selectivity via docking, it was observed that no stable interaction was noted between the compounds and the TRP channel of the predictions.

Further investigations using in silico studies were employed to support the larvicidal mechanism of action of *Siparuna* essential oils against *Aedes aegypti*, aiming to verify which essential mosquito proteins could be molecularly affected by the sesquiterpenes γ-elemene and germacrene D (Montaño-Campaz et al. 2025). Thus, homology modeling of AChE was performed in the Swiss-Model, where high structural identity and model validation were verified with a Ramachandran of 93% of residues in favorable regions and a QMEAN factor of − 1.04, demonstrating that the enzyme structure is robust for docking. The data suggest that sesquiterpenes do not act as classic covalent inhibitors and likely exert non-competitive or mixed inhibition, in addition to interfering with the enzyme's conformational dynamics, leading to acetylcholine accumulation and, conversely, neurotransmission dysregulation, resulting in larval paralysis and death. The study demonstrated that the ligands exhibited van der Waals interactions with important amino acid residues, with germacrene D showing an alkyl-hydrophobic interaction affinity with Phe587 and γ-elemene showing a π-alkyl interaction affinity with Phe587 and Phe591 (Ramachandran and Sasisekharan 1968; Trott and Olson 2010; Waterhouse et al. 2018).

Within the scope of prospecting for compounds with antimicrobial potential, an important study in this section focuses its analyses on molecular docking and molecular dynamics studies, in relation to metabolites such as atractylone, germacrene D and γ-elemene, which showed significant affinity for essential molecular targets, including insect acetylcholinesterase

(AChE) and key enzymes involved in microbial metabolism (Andrade et al. 2013; Nelita et al. 2015). Molecular docking analyses revealed the predominance of stable hydrophobic interactions at the active site of the target proteins, as well as the punctual occurrence of hydrogen bonds with conserved catalytic residues. These results allowed the ranking of the compounds and the selection of the most promising complexes for the molecular dynamics stage. MD simulations demonstrated that the most active molecules form stable complexes with the target protein, exhibiting convergent RMSD and good ligand fit, while RMSF showed reduced flexibility of the active site residues and maintenance of the main intermolecular interactions over time. Free energy calculations by MM-PBSA confirmed favorable binding affinities, with a predominant contribution from hydrophobic interactions, corroborating the proposed antimicrobial mechanism. These interaction patterns suggest plausible mechanisms of enzymatic inhibition, corroborating the effects observed in in vitro and in vivo biological assays previously described in the literature (Melo et al. 2017; Santana de Oliveira et al. 2020).

In the neurobiological field, Martins et al. (Martins et al. 2021) conducted computational screenings targeting acetylcholinesterase (AChE) inhibition, which indicated that oxygenated sesquiterpenes, especially shyobunone and iso-shyobunone, showed satisfactory steric complementarity with the binding site and peripheral regions of the enzyme, with the latter being the strongest ligand as it presented the best binding energy (-8.0 kcal/mol). Docking of the two binding sites revealed two preferential pockets in the AChE catalytic channel. The innermost region, A, involves residues Trp108, Tyr146, Gly143-148, Try155, Phe315, Tyr355, and His494. This region proved to be the closest, but not coincident with, the classic catalytic triad (Ser-His-Glu). The outermost region, site B, peripheral to the enzyme, involves residues such as Tyr94, Trp304, Phe313-315, Tyr355, and Tyr359, which is associated with allosteric modulation of AChE. These findings support the hypothesis of a possible non-competitive inhibition mechanism, previously suggested by kinetic and spectroscopic analyses, and also showed dominant hydrophobic interactions and some stabilizing hydrogen bonds (Valentini et al. 2010; Andrade et al. 2015).

Conclusions

This integrated scientometric, chemometric, and pharmacological analysis demonstrates that *Siparuna guianensis* essential oil is a chemically polymorphic natural product whose biological performance is strongly chemotype-dependent. Current evidence supports consistent bioactivity in insecticidal, antiparasitic, antimicrobial, enzyme-inhibitory, and antitumoral models, yet the field remains largely driven by chemical profiling and preliminary screening assays. Mechanistic pharmacology, dose, response characterization, and translational validation are still limited. Future research should prioritize chemotype standardization linked to botanical origin, plant organ, and extraction parameters to improve reproducibility across studies. Mechanism-oriented investigations using molecular, cellular, and systems pharmacology approaches are needed to confirm proposed targets such as acetylcholinesterase, membrane integrity pathways, and replication enzymes. Comparative testing of isolated major constituents versus whole-oil matrices should be expanded to clarify synergistic and antagonistic interactions. In vivo pharmacokinetics, toxicology, and selectivity studies in mammalian models represent a critical gap that must be addressed before therapeutic consideration. Advanced delivery systems, including nanoformulations and encapsulation platforms, show promise for enhancing stability, bioavailability, and safety and should be systematically optimized. Finally, multicenter collaborative studies integrating metabolomics, chemometrics, and pharmacodynamics are recommended to establish predictive chemotype-activity models and accelerate the rational pharmacological development of *S. guianensis* essential oil.

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J.J.S.M.; M.S.O.; Writing-review & editing: M.S.O.; E.A.F.J.; C.S.F.M.; K.S.S.V.; R.K.; Supervision: M.S.O.; Project administration: M.S.O.; Funding acquisition: M.S.O.; E.A.F.J.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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